ASSIGNMENT 1 OF 2: NETWORK FUNDAMENTALS AND SUBNETTING DESIGN FOR BLUEWAVE INDUSTRIES LTD.

DECEMBER 9, 2025

STUDENT ID : UE22116
GUSTO College

Table of Contents

Introduction....4
Reporting the Responsibilities for BlueWave Industries Ltd.,....4
Types of networks and their suitability to businesses....4
Local Area Network (LAN)....4
Wide Area Network (WAN)....5
Wireless Local Area Network (WLAN)....5
Metropolitan Area Network (MAN)....6
Personal Area Network (PAN)....6
Network Standards....7
IEEE 802.3 (Ethernet Standard)....7
IEEE 802.11 (Wi-Fi Standard)....8
TCP/IP Standard Suite....8
IEEE 802.1Q (VLAN Standard)....8
Network Topology Analysis and Justification....9
Star Topology....9
Mesh Topology....10
Bus Topology....12
Ring Topology....13
Hybrid Topology....15
BlueWave Industries Topology Recommendation....16
Summary of Impact: Topology, Communication and Bandwidth....16
Explanation of the role of key networking protocols and how they support communication....17
Transmission Control Protocol/Internet Protocol - TCP/IP....17
Dynamic Host configuration Protocol - DHCP....18
Domain Name System - DNS....19
Hypertext Transfer Protocol/ Hypertext Transfer Protocol Secure – HTTP/HTTPS 20
File Transfer Protocol - FTP....21
Simple Network Management Protocol - SNMP....22

Simple Mail Transfer protocol - SMTP 23
Secure Shell Protocol - SSH 24
Network Time Protocol - NTP 25
Address Resolution Protocol – ARP 27
OSI and TCP/IP Models Analysis 28
The OSI Reference Model 28
The TCP/IP Model 30
Explanation of protocols and devices map to different layers 32
Protocol and Device Mapping 32
Application to BlueWave Industries 33
Assessment of Networking Principles and Protocol Effectiveness 34
Implementation of Core Networking Principles 34
Protocol Synergy Enabling Network Effectiveness 34
Security Implementation and Quality of Service 35
Scalability and Long-Term Effectiveness 36
Discussion Networking Devices and Operating Principles 36
Switches 36
Routers 37
Wireless Access Points 38
Firewalls 38
Server Types 39
Discussion the interdependence of workstation hardware and relevant networking software 40
Hardware Parts used in workstations 40
Networking Software Components 40
Interdependence Analysis 40
Designing an IP Addressing Scheme and Subnetting Structure for BlueWave Industries Ltd. 42
IP Addressing Requirement Analysis 42
Organizational Structure and Departmental Overview 42

Connection Network Devices Analysis 43
Server Infrastructure Requirements and Justification 43
Servers of the IT Support Department (5 servers) 43
Servers (2 servers) of the finance department 44
Other Departmental Servers (4 servers 45
Present and Future Host Requirement 45
Summary of IP Addressing Requirement 46
Purpose and Importance of Subnetting 46
Security Advantages Network Segmentation 47
Traffic Control and Performance Maximum 48
Scalability 48
Development of an appropriate private IP addressing scheme 49
Selection and Justification of IP Range 49
Subnetting VLSM Methodology 50
Subnetting Calculations 51
For Sales & Marketing Department 51
For Administration Department 52
For Finance Department 53
For Human Resources Department 54
For Management & Executive Offices Department 56
Calculation Summary and Validation 57
Implementation Considerations 58
Using critical reflection to evaluate the network system, security requirements and quality of services needed for enhance the valid conclusion 59
Network System Evaluation 59
Security Requirements Assessment 59
Quality of Service Analysis 60
The Valid Conclusions 60
Conclusion 61
References 62

Introduction

My name is Thet Htoo Zin, and I work as a Junior Network Technician at "TechNova" Network Solutions, a firm that deals in the services of the network configuration and IT infrastructure. This report is a network design proposal of our client, BlueWave Industries Ltd., whose office activities are growing and it needs good IP addressing and subnetting scheme.

BlueWave industries Ltd. has six departments namely: Administration (18 users), Human Resources (10 users), Finance (12 users), Sales and Marketing (25 users), IT Support (8 users) and Management and Executive offices (6 users). It should also have the capability of supporting shared resources such as printers, servers, voip phones, and wireless access points and have the capacity to expand in future.

This report has been divided into two sections Part I explores networking principles, such as types of networks, network topologies, and protocols that will be used in business contexts whereas part II elaborates a workable subnetting design together with the specific calculations of each department. This report ends with critical analysis of the network system, security requirements and quality of service factors.

Reporting the Responsibilities for BlueWave Industries Ltd.,

Types of networks and their suitability to businesses

Local Area Network (LAN)

Local Area Network (LAN) is a network architecture where devices are networked over a small geographical location, usually in a single building or a single campus. In the case of BlueWave industries Ltd., the LAN technology has major advantages such as high-speed connection, affordability and centralized

management of the network resources. A LAN infrastructure could be used to interconnect all the six departments of the organization and help in sharing printers, servers and other network resources with ease. Nevertheless, LANs have limitations, which concern geographical coverage limitations and necessitating the physical investment in infrastructure. Although such limitations exist, LAN technology is very appropriate to the present operational design of BlueWave Industries as it is fast and reliable to support day-to-day business operations in a single office facility. (GeeksforGeeks, 2025)

Wide Area Network (WAN)

Wide Area Networks (WAN) are used to extend the coverage of connectivity over local areas to cover various geographical boundaries across the cities, countries or continents. The first advantage of WAN technology is that it allows expanding business as well as providing an opportunity to access organizational resources remotely. In the case of BlueWave Industries, WAN implementation would be mandatory in case the company resolves to open branch offices in other cities or regions. Nonetheless, WAN solutions have limitations such as increased cost of operation, low speed in transmitting data as compared to LANs, reliance on third party services, and greater vulnerability in security. Although this is not a pressing concern in the single location operations of BlueWave at the moment, WAN infrastructure must be factored off in the network design to support future plans of the company expansion. (Kapoor, 2025)

Wireless Local Area Network (WLAN)

The WLAN or wireless Local Area Networks are wireless networks that are not connected by cables and they rely on radio frequency technology to facilitate mobility of devices within the coverage radius. WLANs have a high value to the contemporary business setting such as mobility of users, flexible connection of devices, ease of installation, and compatibility with various mobile devices. In case of BlueWave Industries, it is necessary to implement WLAN via wireless access

points to support the increased number of mobile employees, provide connectivity in meeting rooms and free zones, and support the policy of Bring Your Own Devices (BYOD). The weaknesses of WLAN technology are that it has security weaknesses whereby it should be used with sound encryption protocols, there is a possibility of radio frequency interference, limitation in the range of coverage as well as sharing of bandwidth with other devices connected to it. Nevertheless, the integration of WLAN is important in the network design of BlueWave to address the current expectations in the working environment. (Fritzler, 2024)

Metropolitan Area Network (MAN)

Metropolitan Area Networks (MAN) are networks that enable connectivity within an area of a metropolis or a city (usually 5-50 kilometers). MAN technology provides a higher speed of transmission of data than WANs and has a greater coverage than LANs hence it is ideal in organizations that have more than one building or one campus in the same city. In the case of BlueWave Industries, MAN implementation would not be applicable to the organization unless it has to expand to various office locations within the same metropolitan area. Its limitations are high cost of infrastructure, reliance on municipal or service provider authorizations and limited usability outside cities. The MAN technology is currently not of much value to the immediate operation of BlueWave but is still an option to be considered in the future expansion plans of the organization by going to urban places. (IPCisco, n.d.)

Personal Area Network (PAN)

Personal Area Networks (PAN) provide short range connectivity of personal devices, which is usually limited to a range of 10 meters and uses technology such as Bluetooth or infrared. The advantages of PANs include: easy connectivity of peripherals wirelessly, low power, and easy process of pairing of the devices. PANs however have very big limitations such as very limited range, small bandwidth capacity and a small number of simultaneous connections. In the case of BlueWave Industries, PAN technology is not heavily utilized in the organizational environment, rather it is used to serve the personal needs of employees like connecting Bluetooth headsets, wireless keyboards, mice and personal device synchronization. PAN

technology is not a significant aspect in the design of the entire network infrastructure in the institution. (Anon., 2025)

Network Standards

Network standards provide technical requirements that guarantee interoperability, compatibility, and uniform performance of networking equipment of various manufacturers. These are the requirements that BlueWave Industries needs to make sure that it has a reliable connectivity and future-proof investment of infrastructure.

IEEE 802.3 (Ethernet Standard)

The wired Ethernet technology that is defined by the IEEE 802.3 standard can work at different speeds such as 100 Mbps (Fast Ethernet), 1 Gbps (Gigabit Ethernet), and 10 Gbps (10-Gigabit Ethernet). In the case of BlueWave industries, Gigabit Ethernet (1000BASE-T) offers the best features such as high-speed connectivity to the 79 people using it, support of bandwidth applications, affordability of implementation using either cat5e or cat6 cable and adequate capacity to support files sharing and voice over internet connections. The limitations are the cable length of 100 meters when using twisted pair cabling, the necessity to have a good cabling infrastructure, and possibility of having to upgrade to fiber optic cabling in the future in case the 10 Gbps needs to be upgraded. This is the best standard to be used in connecting all the six departments using switched Ethernet infrastructure to offer the performance and reliability required in the critical business operations. (Committee, 2024)

IEEE 802.11 (Wi-Fi Standard)

The wireless networking technology is regulated by a family of standards known as IEEE 802.11, which now includes 802.11n (Wi-Fi 4), 802.11ac (Wi-Fi 5) and 802.11ax (Wi-Fi 6). In case of BlueWave Industries, the application of the 802.11ac or 802.11ax standards will provide such advantages as wireless working position to its employees, ability to support mobile devices and BYOD policies, flexibility in meeting areas, as well as 1-3 Gbps speed with the latest equipment. Security vulnerability (needs WPA3 encryption), the possibility of interference with the neighboring network, the number of simultaneous connections available to each access point (usually 30-50 devices), performance problems at a distance, and physical barriers are constraints. This is one of the requirements of the wireless access points that BlueWave needs to have to serve the needs of mobile workforce and provide flexibility in workspace configurations.

TCP/IP Standard Suite

TCP/IP is the universal standard protocol stack of internet and in-house network communication. The advantages to the BlueWave Industries are that it is universal to all devices and operating systems, it has been proved to be reliable and scalable within networks of any size, it has full protocol stack with every business application and its vendor support has made it viable in the long run. The biggest limitation is complexity with the need to have trained IT specialists to do the appropriate configuration, troubleshooting and security application. TCP/IP is the backbone of the entire network structure of BlueWave and the subnetting plan that needs to be used to divide the six departments and still ensure the inter-departmental communication is controlled.

IEEE 802.1Q (VLAN Standard)

Virtual LAN (VLAN) can be implemented with the help of the IEEE 802.1Q standard that provides the possibility to logically divide the physical infrastructure.

To its advantage, it offers increased security with isolation of traffic between departments, increases performance by reducing broadcast domains, allows flexibility in network administration with no recabling (physical) and Quality of Service (QoS) prioritization support. Limitations are that it needs VLAN capable managed switches, more difficult to configure than a flat network and there is a likelihood of misconfiguration errors that might cause connectivity disruption. This standard plays a vital role in ensuring that BlueWave Industries adopts the necessary departmental network segmentation to enable the Administration department, HR department, Finance department, Sales and Marketing department, IT Support department as well as the Management department to be separated Layer 2 segments but share the same switching infrastructure. (Committee, 2024)

Network Topology Analysis and Justification

Star Topology

Star topology is a type of network structure that involves all devices being connected to a central networking device usually a switch or a hub. The topology has many benefits in a business setting such as, an easier way of troubleshooting the network as the issues can be limited to one connection, isolated fault domains where the failure of one device does not impact the whole network, easy addition of devices as all that is needed is a cable connection to the central switch, and easy management capabilities as a problem can be easily monitored and controlled.

Communication is greatly affected positively because the star topology with the application of modern switches ensures each connection has dedicated bandwidth that removes the collision domains and ensures equal performance at all times despite the network load. Star topology however has drawbacks such as the possibility of having the single point of failure at the center device although this is reduced by redundant switch topology, but it also has higher cabling requirements than bus topology because each device must have its own cable run to the center. The bandwidth demand is effectively controlled using the new switched Ethernet

technology that offers full-duplex communications capable of supporting transmission and reception at the same time in addition to allocating exclusive bandwidth to individual devices connected and guaranteeing 1 Gbps connection per port. (Cox, 2022)

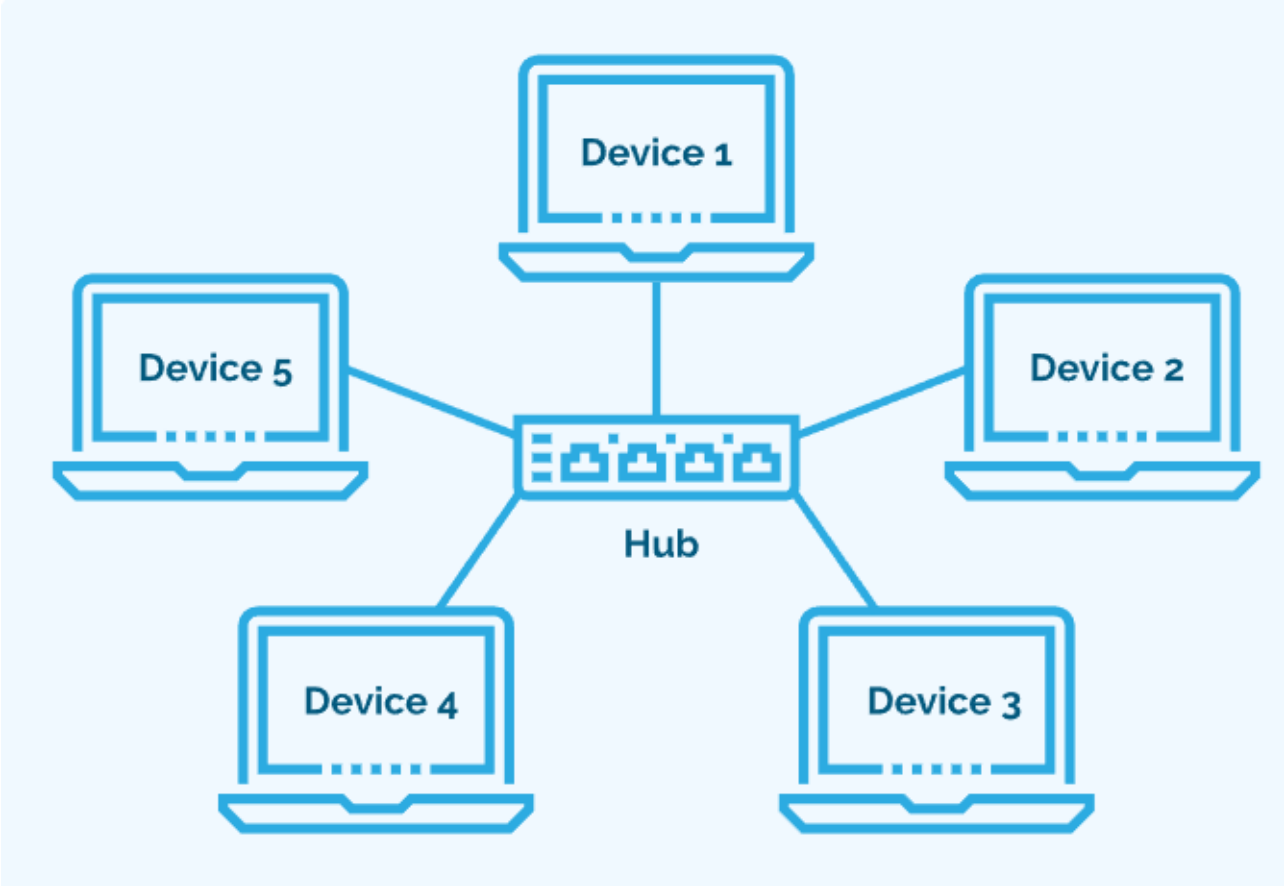


Figure (1) – Star Topology

Mesh Topology

Mesh topology is used to form a topology in which each device has direct links to all devices to give a network with several redundant data transmission routes. The benefits of this set-up are superb features such as maximum redundancy which guarantees the resilience of the network, removing single points of failures by providing multiple alternative paths, high fault tolerance capability and situation to

continue with operation even when linkages fail, and optimal distribution of loads across multiple paths.

Communication impact is extremely positive and there are various alternative routes that would provide constant connectivity even when there was a failure of individual links and automatically reroutes traffic around the congested or failed links. Mesh topology however has major implementation limitations to business, rather expensive cabling costs which grow exponentially with the number of devices, elaborate installation processes that demand planning and documentation, difficult to manage maintenance processes and large switch port requirements. The bandwidth effect is also mostly positive, since traffic is spread across a number of paths, and one link will not overload. Although it has the advantage of reliability, full mesh topology is generally not a viable option in BlueWave Industries because of the costs and complexity factors, although partial mesh topology might be a useful choice in high stakes infrastructure connections. (Cox, 2022)

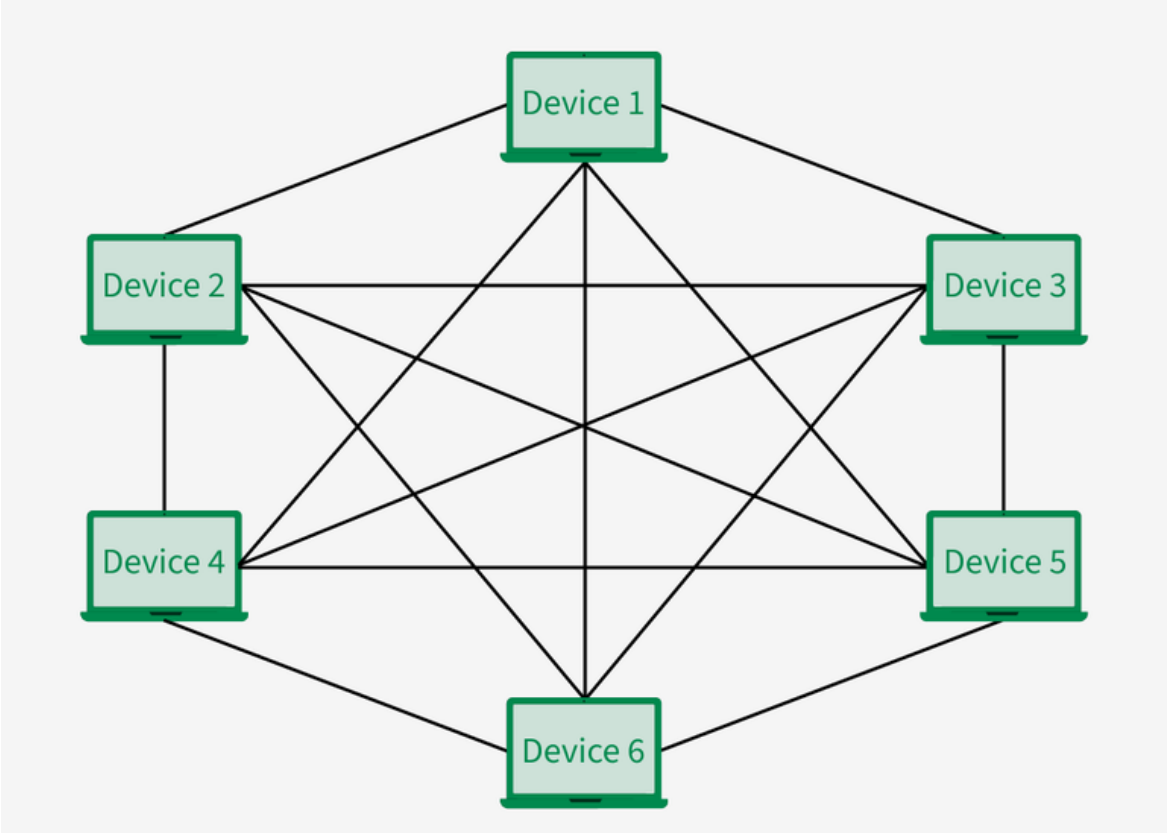


Figure (2) – Mesh Topology

Bus Topology

Bus topology makes use of a single communication line that all the devices are connected to share bandwidth over a common medium. Its main strengths are easy installation processes which only need a few cables, affordability when implementing a small network and its ability to extend the network with the addition of devices to the bus. Nevertheless, bus topology has significant drawbacks which render it inappropriate in the contemporary business networks.

The common communication channel also forms a single point of failure whereby failure on the main cable affects the whole network. As the number of devices increases, performance is decreased and the higher the rate of collision, the less effective the bandwidth and throughput are. This has an adverse effect on communication as the common bandwidth causes contention between devices within a larger implementation where the CSMA/CD (Carrier Sense Multiple Access with Collision Detection) contention mechanisms are responsible of adding delays. It is not an easy task when it comes to the troubleshooting because the problems may be encountered on a network segment. Bus topology is the old technology that is not suitable to the needs of BlueWave Industries due to reliable and high-performing connectivity among the six departments with 79 users and several shared resources. (Cox, 2022)

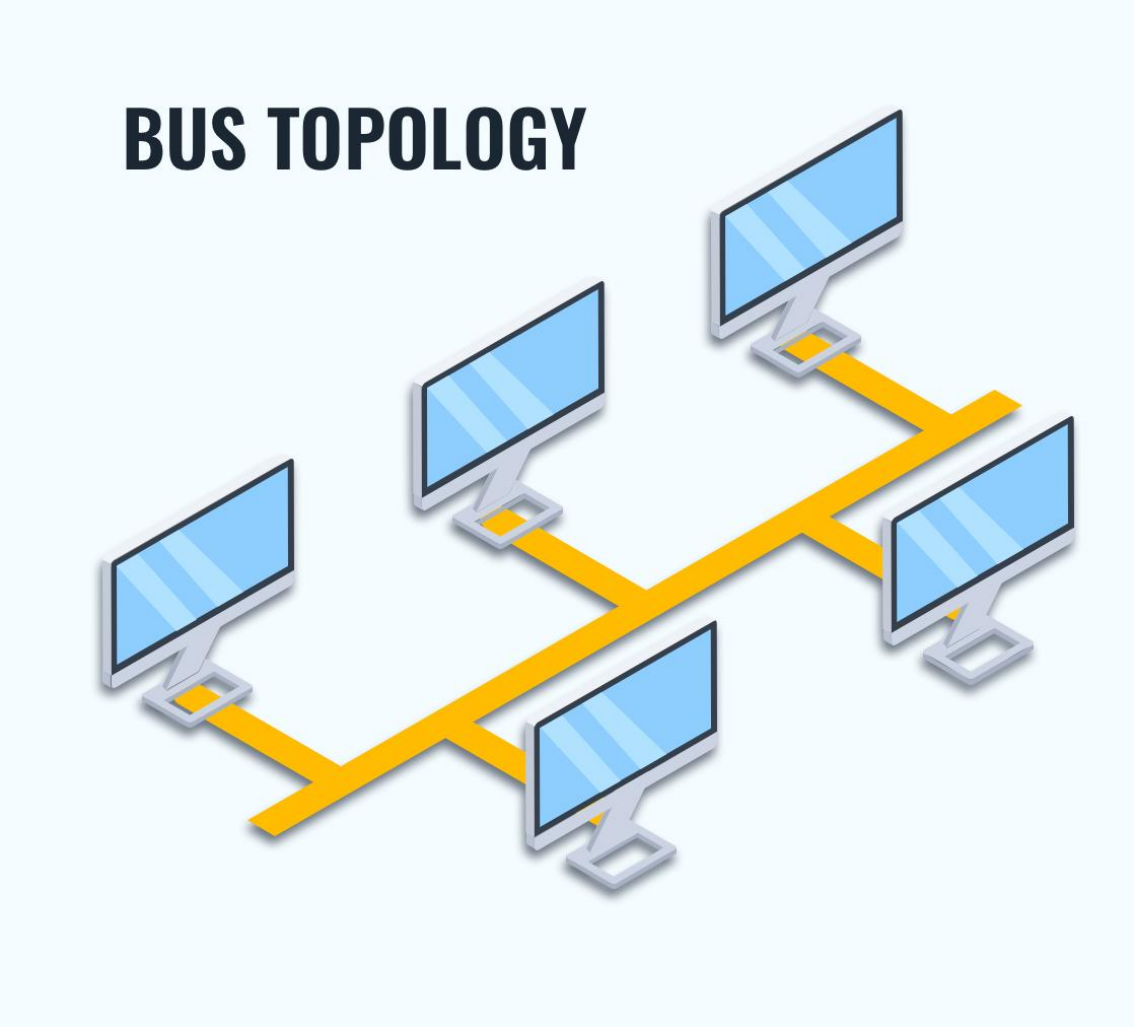


Figure (3) – Bus Topology

Ring Topology

Ring topology is a type of topology where devices are interconnected in a circular distribution with data flowing in either direction or both directions around. Such a topology is characterized by high accessibility to all devices with equal consideration of the opportunities and predictability of the nature of performance and navigable organization of the data flow with token-passing procedures that inhibit the occurrence of overlaps. Past implementations of this topology include Token Ring and FDDI (Fiber Distributed Data Interface).

Nevertheless, ring topology has some key drawbacks such as network failure whenever a single ring in the ring is broken, unless dual ring topology is taken, it is hard to add or remove devices, which involves network interruption and its troubleshooting is not very easy. The bandwidth effect exhibits predictable trends using a controlled access protocol that eliminates the possibilities of collisions, but may introduce latency when devices are busy sharing the network and need to wait before the token is available. Such drawbacks as historical status of technology, and the lack of support of modern vendors make ring topology an inappropriate choice in the case of BlueWave Industries requiring flexible and easily scaled network infrastructure that could support modern business demands. (Cox, 2022)

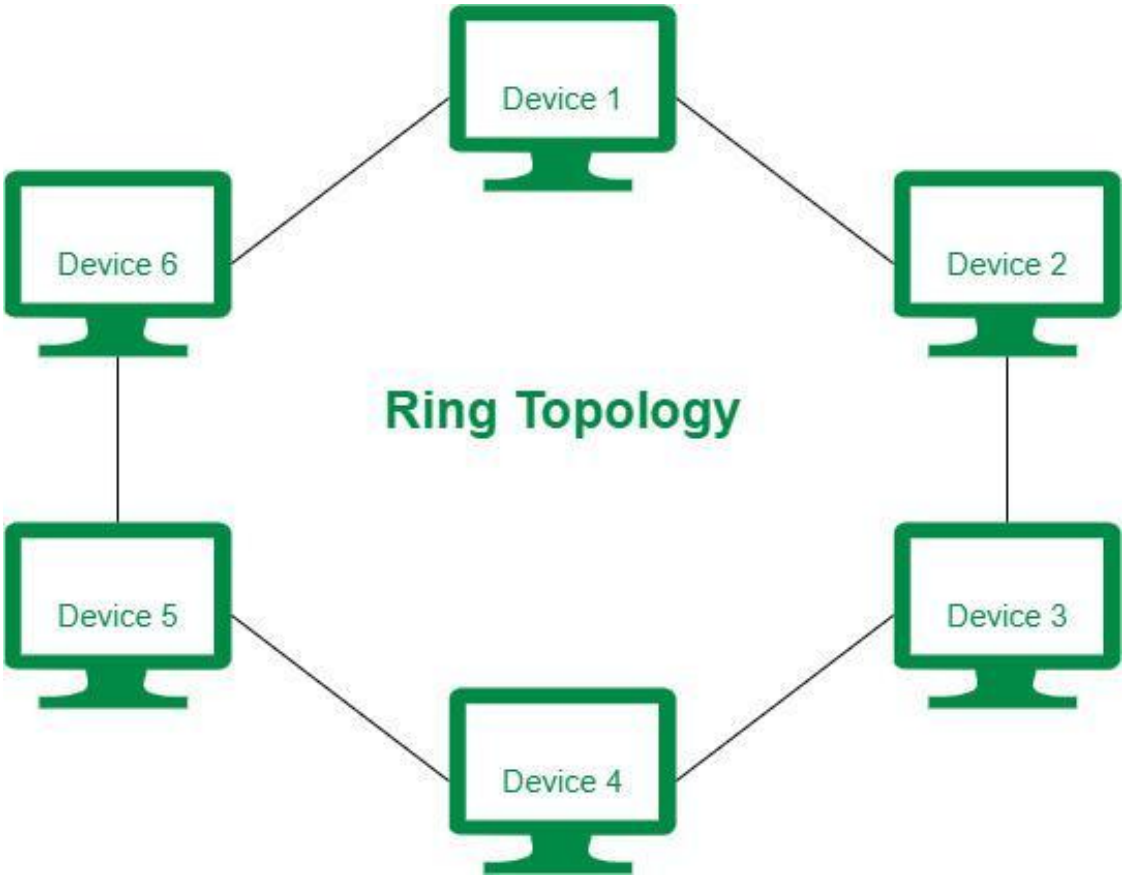


Figure (4) – Ring Topology

Hybrid Topology

Hybrid topology is a topology where two or more types of basic topology are used to take advantage of their respective strengths, and overcome the shortcomings of each. This solution provides outstanding flexibility to support custom designs to fit particular organizational needs, scaling to support expansion without basic redesign, and reliability through a variety of topology deployment in network segments.

The most widespread hybrid solution is a mixture of star topology and tree topology that form hierarchical network organizations where the access layers are arranged in the form of a star and are linked by a backbone infrastructure. This design style is favorable to the multi-floor or multi-building expansion of BlueWave Industries. Although hybrid topology involves more design complexity that would need experienced planning and documentation, and high cost, as opposed to single topology implementation, it offers the best solutions to a wide range of organizational needs. Its effect on communication and bandwidth also depends on the combination of topology in use though it tends to be cost-effective, performance, and reliability-wise. (Cox, 2022)

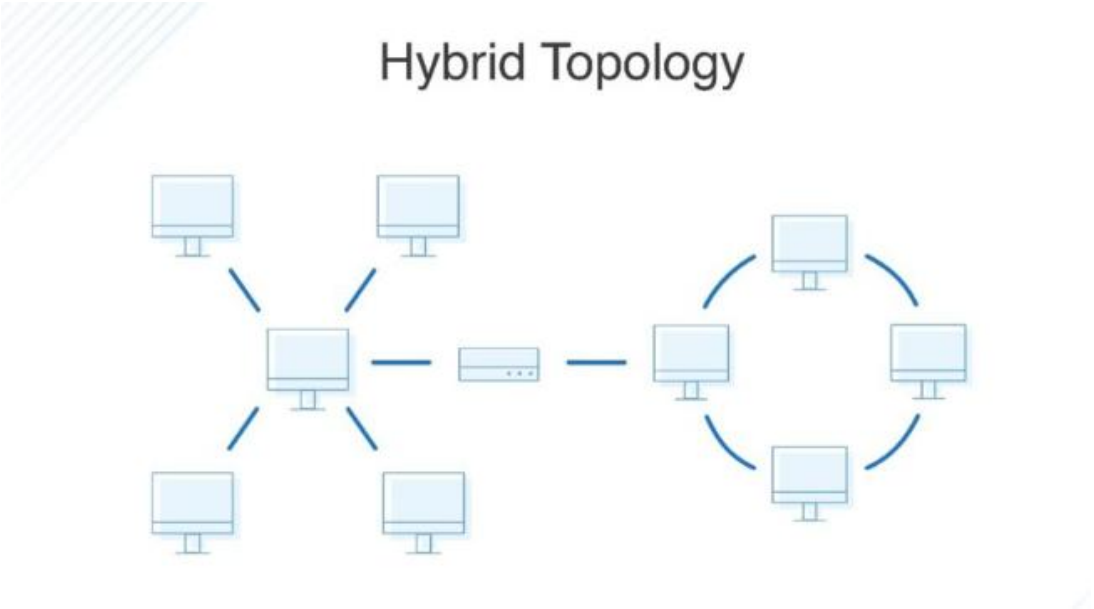


Figure (5) – Hybrid Topology

BlueWave Industries Topology Recommendation

Through thorough analysis, the star topology would be the best option to use at BlueWave Industries Ltd. The following recommendation can be explained by a number of essential aspects that meet the unique needs of the organization. First, star topology will allow centralized management by means of the eight-member IT Support Department, which will make monitoring, troubleshooting and maintenance operation easy due to one-point management. Second, the topology facilitates simple scalability to cater to the future growth, and this is in line with the growth trend of BlueWave, and the need to support more users and devices than the present 79 users. Third, star configuration eases security segmentation by use of VLAN implementation, each of the six departments, Administration, HR, Finance, Sales and marketing, IT Support, and Management, can enjoy physical infrastructure and still enjoy security isolation. Fourth, the isolated fault domains help in ensuring that a fault in one department does not propagate to other areas and keep the business running and the extent of the disruption is minimized.

Lastly, topology star offers the dedicated bandwidth which can serve the 79 users of the organization as well as shared resources such as printers, servers, VoIP phones, and wireless access points to ensure steady performance of all the devices connected. (IPCisco, n.d.)

Summary of Impact: Topology, Communication and Bandwidth.

Topology, Communication and Bandwidth. The chosen star topology has a number of paramount effects on the efficiency and bandwidth needs of the communication in BlueWave Industries. Communication wise, star topology gets rid of the areas of collisions in bus topology, which allows simultaneous full dux communication among all the devices and the central switch, which is necessary to support 79 simultaneous users in six departments. The dedicated switch-to-device connections are chosen with the purpose that the communication between the Finance Department and the financial servers would be independent of Sales and

Marketing Department activities and would not interfere with each other. Optimized bandwidth is achieved by dedicated switch ports with 1 Gbps connections to each of the user devices where the connections add up to the central routing infrastructure, with higher capacity uplinks of 10 Gbps or higher. (IPCisco, n.d.)

This topology option will guarantee that sales and marketing department activities with bandwidths of 25 users including video conferencing and large files transfers do not affect financial department which has 12 users accessing financial applications and databases. The centralized architecture also enables monitoring bandwidth to be managed using switch management interfaces, Quality of service (QoS) implementation, which enables priority in assigning VoIP traffic to support the VoIP phones and important business applications of an organization over the general web browsing and non-crucial traffic. Since the BlueWave industries has a growing number of workers and devices, the star topology can support a greater bandwidth requirement without necessitating a fundamental redesign of the topology or large-scale recabling initiatives, quite simple upgrades to the switch will suffice to support the growing bandwidth requirements. (IPCisco, n.d.)

Explanation of the role of key networking protocols and how they support communication

Transmission Control Protocol/Internet Protocol - TCP/IP

TCP/IP is the basic protocol stack on which the current network communication and internet connectivity rely. It is a stack protocol and it can work at several levels, at a high level there is the IP which offers logical addressing based on 32 bits IPv4 addresses and routing facilities that directs packets within networks and at the low level there is TCP which offers reliability as a connection-oriented data delivery protocol, error detection and retransmission and flow control.

In the case of BlueWave Industries, all the communication in the network is based on TCP/IP, which is used to facilitate communication between the six departments and the subnet structure that supports the security segmentation

needs. The protocol functions in aiding the communication process fully and includes addressing of all the 79 user devices and servers and network infrastructure, routing of packets which directs the traffic that exists between departmental subnets, error detection and correction which maintains the data integrity of financial and business critical data and end to end connectivity which is provided by the three-way handshake mechanism. The universal nature of TCP/IP also means that it can support all network devices, all operating systems such as Windows, MacOS, and Linux, and all applications needed to support the functions of BlueWave, web browsers to specialty business applications. (Anon., 2023)



Figure (6) - TCP/IP

Dynamic Host configuration Protocol - DHCP

Drastically, it is through DHCP that the IP address assignment process can be automated, so they only require manual configuration and less overhead due to automated assignment. The DHCP servers have tables of free IP addresses per departmental subnet and issue addresses to devices dynamically on network connection in a four-step process Discovery, Offer, Request and Acknowledgement. In the case of BlueWave industries where the number of users is 79, spread among six departments and other network devices such as printers, VoIP phones and wireless access points, the implementation of DHCP would greatly ease the workload

of the IT Support Department by eliminating the need to handle large numbers of users per department.

The protocol allows IP address conflicts to be avoided by means of centralized control and tracking of leases and shows lease information to be held to accountability and also used in troubleshooting, allows efficient use of IP addresses by temporary assignment and automatic reclamation address, and allows network changes by updating configurations at a central location instead of visiting each machine individually. DHCP facilitates the communication by assuring that all devices are assigned appropriate network configuration parameters such as IP address according to their department subnet, subnet mask to determine network boundary, default gateway to provide internet and inter departmental communications and name resolution by using address of DNS server, which does not require IT Support Department to make each new device active on the network. (Anon., 2023)

Domain Name System - DNS

Domain Name System (DNS) is a critical name server that offers the process of converting a human friendly domain name into an IP address that is used to communicate over the network. DNS is based on hierarchical and distributed database mechanism that allows scalable resolution of names in both the internet and local networks and local DNS server caches frequently requested data in order to reduce the response time.

In the case of BlueWave Industries, DNS will allow easy access to internal servers like “fileserver.bluewave.local” or “email.bluewave.local”, email systems with MX record resolution, and the internet resources without making the employees in the Administration, HR, Finance, Sales and Marketing, IT Support and Management departments memorize the IP addresses of the servers. The protocol promotes communication by ensuring that resources remain reachable despite the underlying IP address changes whenever servers are migrated or reconfigured, supports email

delivery via MX (Mail Exchange) records that direct messages to relevant mail servers, service discovery to applications (that locate resources based on a DNS query), and supports load balancing by delivering many IP addresses to high-availability services. Any redundancy in the implementation of DNS servers should be done in both primary and secondary to have the name resolution services available to all the departments, which would not cause a single point of failure and thus interfere with email and web access. (Anon., 2023)

Hypertext Transfer Protocol/ Hypertext Transfer Protocol Secure – HTTP/HTTPS

The protocols used to support web-based communication and access to web applications are HTTP and its secure version HTTPS. The HTTP is a request-response protocol between the client and servers, which use TCP port 80, and the HTTPS is a protocol that uses Transport Layer Security (TLS), which encrypts data using TCP port 443 in order to protect and authenticate the data transmitted and verify integrity. In the case of BlueWave Industries, HTTPS is essential to the Finance Department dealing with sensitive financial information such as banking institution details and salary, and the Human resources department dealing with confidential employee information such as performance review, employee records.

All these protocols facilitate the communication process by providing access to web-based business applications such as cloud-based CRM systems utilized by Sales and marketing, accounting software utilized by the Finance and HR management platforms, internet resources required by research and business intelligence and API-based integration of various business systems. The use of HTTPS guarantees confidentiality of data resistive to eavesdropping of sensitive data, integrity of data by using cryptographic hash to identify alteration and authentication of the server using cryptographic certificates that ensure validity of the server. (Anon., 2023)

File Transfer Protocol - FTP

File transfer Protocol (FTP) supports the efficient file transfer processes among the network systems. FTP uses a client server architecture, where control connections are made on TCP port 21, and data connections are made on the TCP port 20, in which reliable file transfer is achieved with authentication, ability to navigate directories and choice of transfer mode. In the case of BlueWave Industries, FTP will be used to aid in communication between the departments like Sales and Marketing by allowing them to transfer large file to the Management and to carry out backup of servers where the IT Support Department will transfer important files to backup disks and also provide file access where files will be accessed using username and password authentication.

The Security file transfer properties of SFTP (SSH File transfer protocol) or FTPS (FTP Secure) version are specifically useful in the Finance Department of the company with its secure financial files like payroll data and financial reports that are sensitive and Sales and Marketing Department with 25 users that have a need to share large promotional materials, product catalogues and multimedia content. The FTP functions will also be to assist the IT Support Department in the distribution of software within the company, installation of software and updates in the departmental computers and server administration such as retrieving log files and updating configuration files. (Anon., 2023)

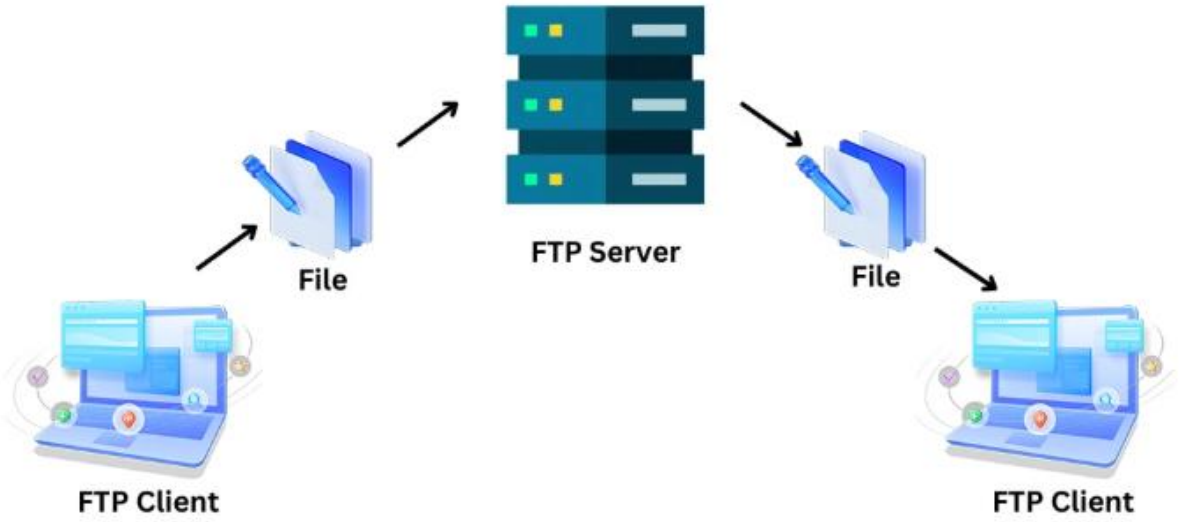


Figure (7) – How FTP protocol works

Simple Network Management Protocol - SNMP

Simple Network Management Protocol (SNMP) offers monitoring and management of the networks that are fundamental in ensuring that the networks remain healthy and functional. The SNMP allows centralization of device status, performance statistics and configuration related data using a manager-agent framework with the agents running on network devices reporting to a central management station over UDP ports 161 and 162. In the case of BlueWave Industries, SNMP will provide the eight technicians in the IT Support Department the ability to actively monitor all network objects such as switches linking the six departments, routers, printers, a shared resource, servers, business applications and data, and wireless access points, all of which utilize WLANs.

The protocol aids in communication by allowing the automatic generation of alerts on faulty conditions like switch port malfunctions or excessive usage of CPU resources on servers, capacity planning trending to learn when network improvements are required to handle increasing user demands, bandwidth usage

monitoring to see which departments are consuming network resources and centralized configuration management to provide consistent settings on devices within infrastructure. Implementing SNMP will be necessary to ensure the high-quality network connectivity demanded by the expanding operations of BlueWave, allowing the IT Support Department to detect and solve the problems before users are affected by them, and make informed decisions regarding the investment of infrastructure to support the expansion in the future. (Anon., 2025)

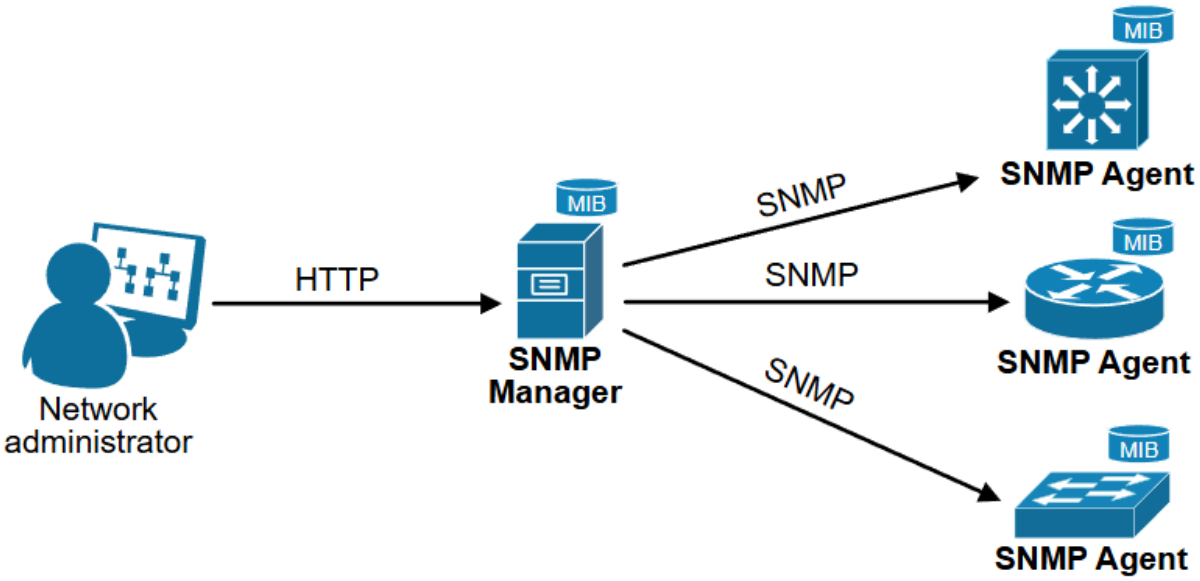


Figure (8) – SNMP Components

Simple Mail Transfer protocol - SMTP

Simple Mail transfer protocol (SMTP) is the protocol that is used to send email messages between mail clients and mail servers as well as mail servers and mail clients. SMTP operates on TCP port 25 for server-to-server communication and ports 587 or 465 for client-to-server encrypted transmission. In the case of BlueWave Industries, SMTP will be fundamental in facilitating email communication to all the six departments, business correspondence, sharing of documents, and coordination between the various departments. The protocol is used along with the DNS MX records to direct email to the correct mail servers and has authentication features

to avoid unauthorized email relay. The Human Resources Department uses it to send out employment offers and policy communications and the Sales and Marketing Department to send out communications to their customers and promotional offers. The Finance Department would also need SMTP to send the invoices and financial reports to the Management. Implementation of SMTP must contain security keys like SPF (Sender Policy Framework) and DKIM (Domain Keys Identified Mail) to discourage email spoofing and to ascertain the veracity of the contents to defend against phishing assaults and save the email status of the BlueWave Industries. (Anon., 2023)

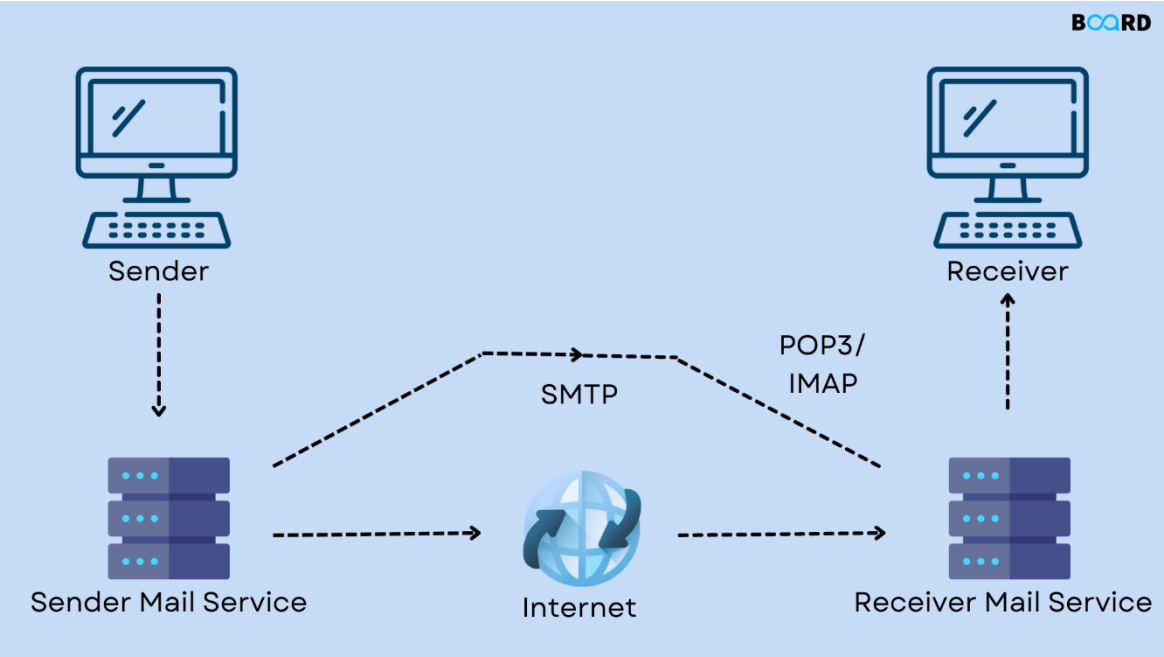


Figure (9) – How SMTP protocol works

Secure Shell Protocol - SSH

Secure Shell (SSH) is an encrypted system to access the network devices and servers remotely, allowing to provide secure administrative control by use of command line and file transfer. SSH is based on TCP port 22 and substitutes the unsafe protocols such as Telnet by encrypting all the communication such as passwords and commands. In the case of the IT Support Department of BlueWave

Industries that has 8 technicians, SSH is essential in providing secure remote control of network switches, routers and Linux/Unix servers without necessarily being physically present in this equipment. SSH has key based authentication that offers better security as compared to password-based authentication, it cannot allow eavesdropping on administrative sessions, it allows secure file transfer via the SCP (Secure Copy Protocol) and SFTP and gives secure tunneling to other protocols. The protocol enables IT Support Department to remediate network problems, implement configuration changes, watch system logs, and carry out maintenance operations remotely without losing information confidentiality and integrity. The implementation of SSH is especially essential when IT employees have to operate the infrastructure during off-peak hours or address the emergency problems without the need to travel to the network equipment locations. (CloudFlare, n.d.)

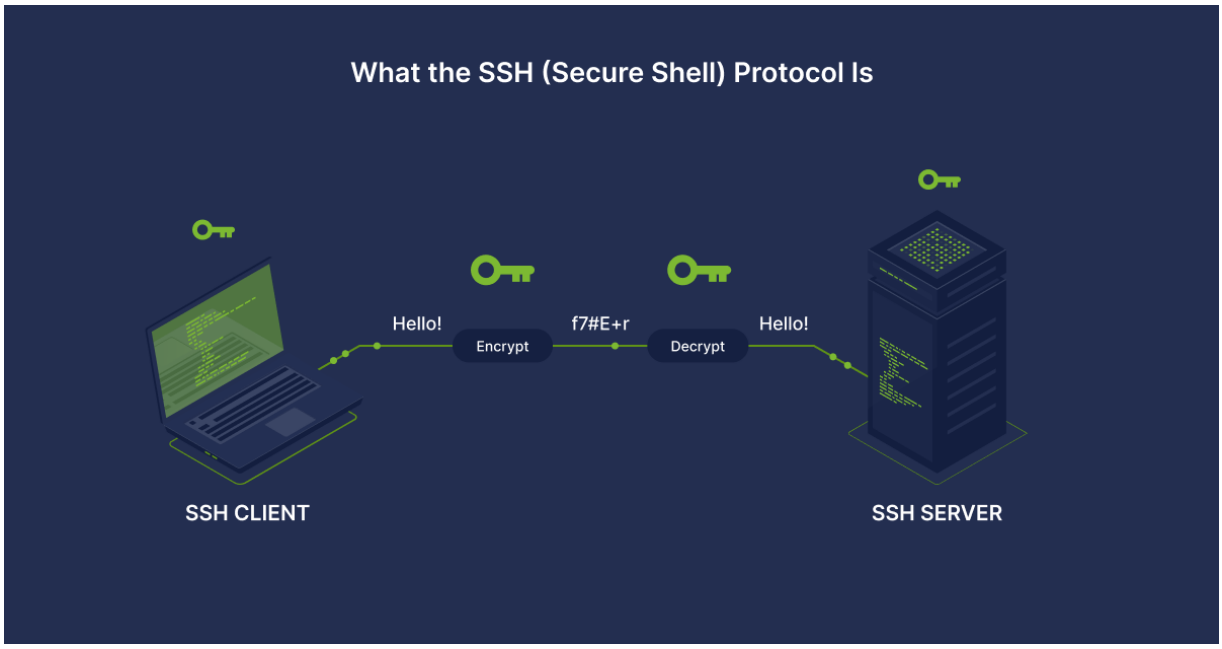


Figure (10) – SSH Protocol

Network Time Protocol - NTP

Network Time Protocol (NTP) coordinates system clocking on devices in the network so that they all maintain an accurate and consistent timekeeping. NTP uses UDP 123 port and can achieve an accuracy of one millisecond by use of the

authoritative sources of time like internet time servers or GPS clocks. To BlueWave Industries, time synchronization would be essential in several business operations such as matching log records between devices as a security incident investigation tool, maintaining precise log timestamps on financial transactions and records in Finance, coordinating the schedule of tasks and backups, and ensuring validating digital certificates with known time limits, and meeting regulatory requirements regarding audit trails.

The protocol accommodates hierarchical time distribution based on stratum levels where stratum 0 model atomic clocks and each successive level receives time input by the previous stratum level. The network of BlueWave must install NTP servers that synchronize to trusted internet time (stratum 2 or 3) and all switches, routers, servers and workstations must be configured as NTP clients keeping synchronized time, which is critical in the errors being solved by the IT Support Department when there is a need to rebuild timelines across the multiple systems. (Anon., 2025)

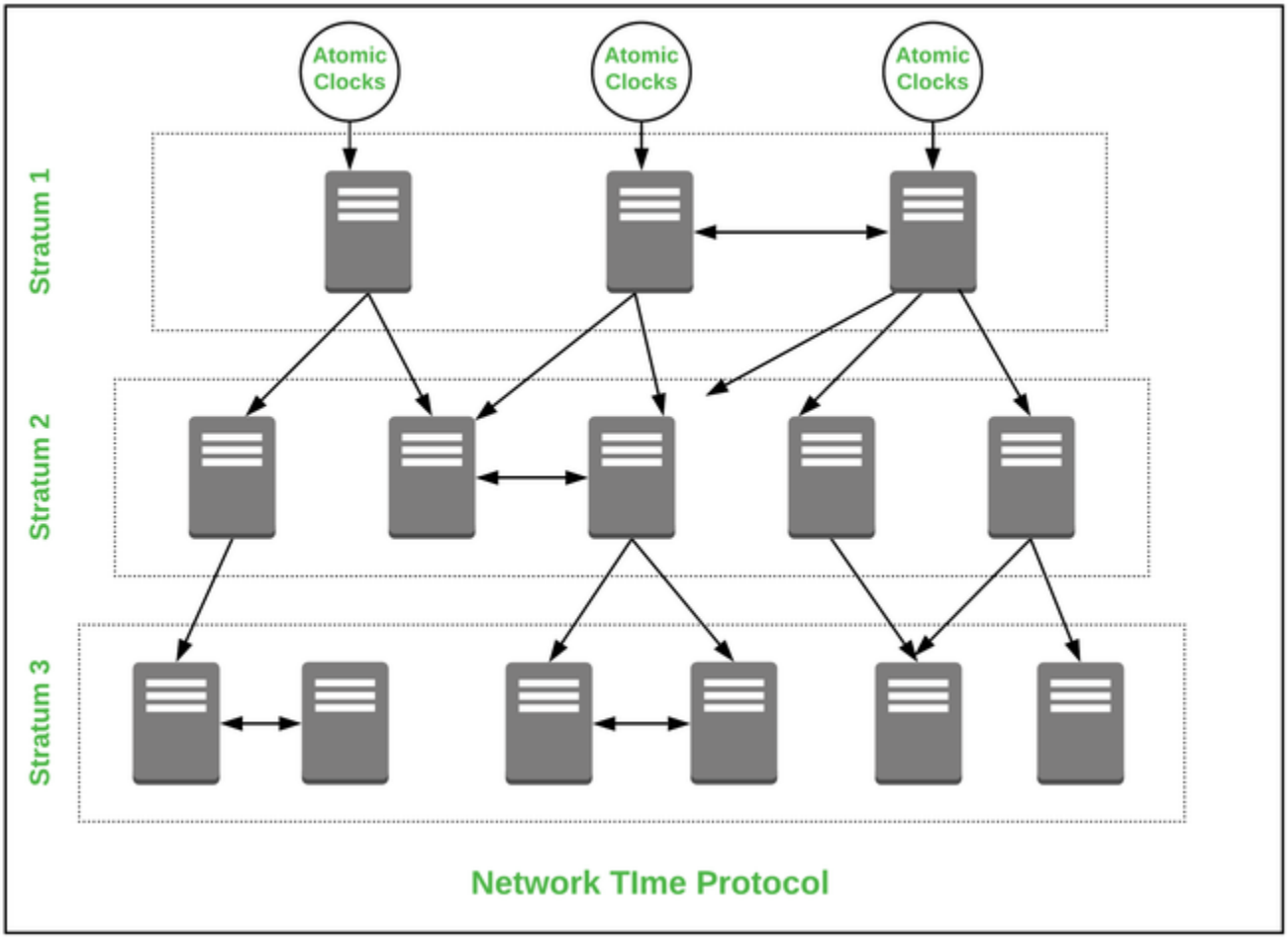


Figure (11) – NTP Protocol

Address Resolution Protocol – ARP

Address Resolution Protocol (ARP) binds IP addresses of the network layer with MAC addresses of data link layer which facilitate communication between segments of a local network. ARP functions at Layer 2 and Layer 3 boundary and keeps ARP cache table in every device containing recent IP-to-MAC address mappings. In the case of BlueWave Industries, ARP is imperative but mostly invisible and it automatically solves the MAC address issues as devices interact within each departmental subnet. When a device requires to transmit information to the IP address on the local network, ARP transmits a request to query Who has this IP address? and the device whose IP matches that one replies with its MAC address and so the switch is able to transmit frames accordingly.

The protocol has implications on network security since the ARP spoofing attacks can reroute any traffic via rogue devices and therefore, ARP inspection functions on the switch are necessary in the confirmation of ARP packets and to avoid spoofing. ARP attacks can be prevented by configuring the critical server in the Finance and HR department with the use of static ARP entries. Anomalies that should be detected by the IT Support Department in the ARP tables are multiple MAC addresses with the same value indicating configuration mistakes or security problems and high ARP traffic indicating network scanning or attacks. ARP is also important to understand how to troubleshoot connectivity problems in VLANs and subnets of departments. OSI/ TCP/IP models Analysis. (Anon., 2025)

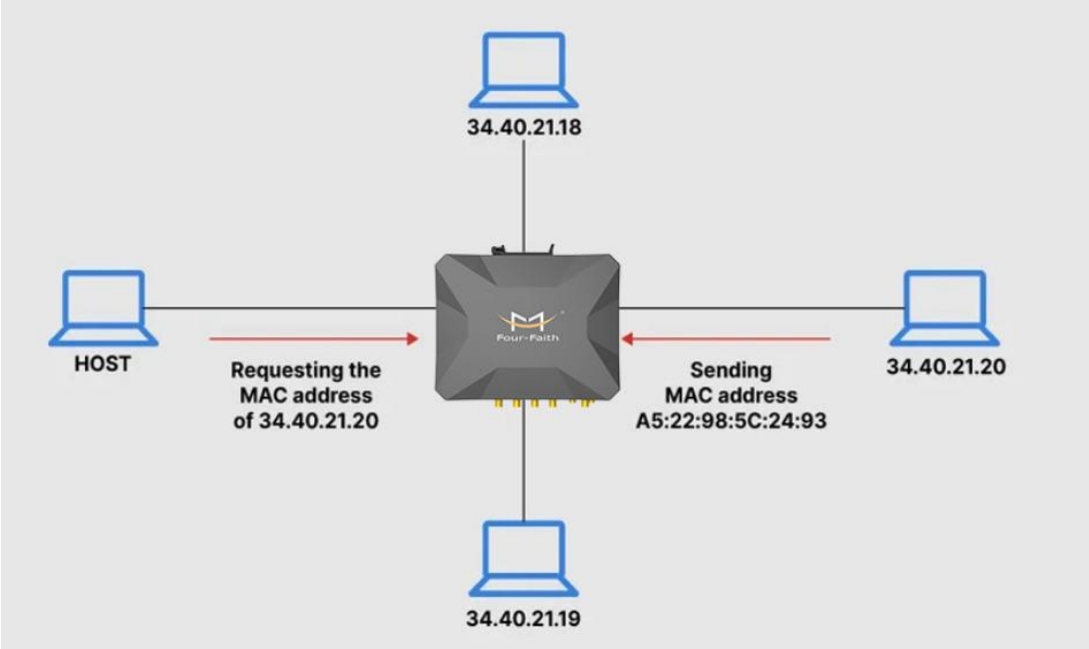


Figure (12) – How Address Resolution Protocol works

OSI and TCP/IP Models Analysis

The OSI Reference Model

The Open Systems Interconnection (OSI) model offers a conceptual framework in the form of 7 layers that can be used to gain insights into the network communication process. Layer 1 (Physical) contains the physical medium of

transmission such as copper cables (Cat5e, Cat6), fiber optic cables, network interface cards (NICs), electromagnetic or optical signaling definition of voltages and light wavelengths. The layer 2 (Data Link) addresses MAC (Media access control) addresses based on 48-bit hardware addresses, frame construction and encapsulation of network layer packets, error control through CRC (Cyclic Redundancy Check) schemes, flow control, and switches mainly operate at this layer based on MAC address tables to make forwarding decisions.

Layer 3 (Network) takes care of logical addressing by IPv4 or IPv6 addresses and routing capabilities that allow the optimum paths across networks, whereas the routers in this layer analyze destination IP addresses to send packets across networks or subnets. Transport Layer 4 (Transport) provides the use of reliable data delivery over TCP with the following features: connection-establishment, acknowledgements, and retransmission features or connectionless transmission over UDP where time-sensitive applications like the VoIP are being used and speed is more important than reliability. The layer 5 (Session) creates, supports, and removes communication sessions between applications and controls dialogs and synchronization. Layers 6 (Presentation) manipulates the data format such as encodings of characters (ASCII, Unicode), encryption and decryption to transmit data in a secure state, and compression to make good use of the bandwidth. The Application (Layer 7) is a layer that offers network services to the end-user applications such as HTTP to access web, FTP to exchange files, DNS to resolve names, and DHCP to configure addresses. (Anon., 2025)

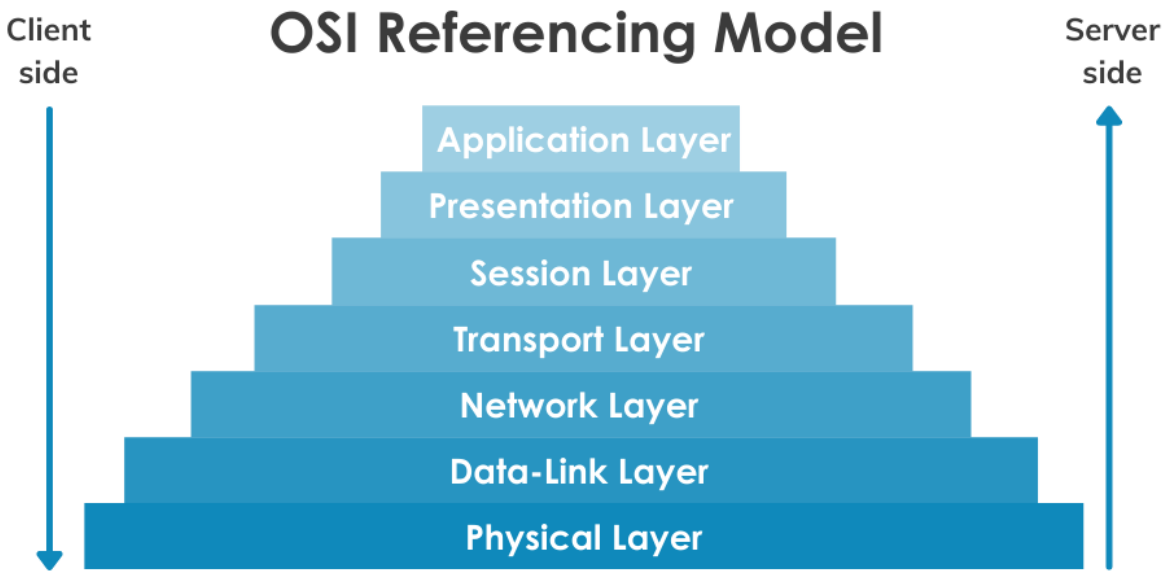


Figure (13) - OSI Referencing Model

The TCP/IP Model

The TCP/IP model further breaks down the OSI model into four operational layers more realistic in protocol implementation. The Network Access layer is an integration of OSI layer 1 and 2, which provides physical transmission using either electrical signaling or radio waves, as well as data link protocols such as Ethernet frame formatting, MAC addressing and media access control schemes based on wired (IEEE 802.3) or wireless (IEEE 802.11) technology.

The Internet layer is the same as OSI Layer 3, where the IP addresses identically label devices in networks and routing, which is the use of IPv4 to determine the current addressing, IPv6 to satisfy the future addressing, ICMP to give diagnostic messages used by ping and traceroute utilities, and ARP (Address Resolution Protocol) to associate IP addresses and MAC addresses. The Transport layer is equivalent to the OSI Layer 4, and offers TCP which is a reliable transmission with connection setup, ordered delivery and flow control that is useful in file transfers and email as well as UDP which is a connectionless communication that has minimal overhead and because of this is suitable in VoIP, video streaming and DNS queries. The Application layer is a combination of OSI layers 5, 6 and 7, it contains all protocols that directly relate to user application such as HTTP and HTTPS, file

transfer over FTP, email transfer over SMTP, name resolution over DNS, address configuration over DHCP and network management over SNMP.

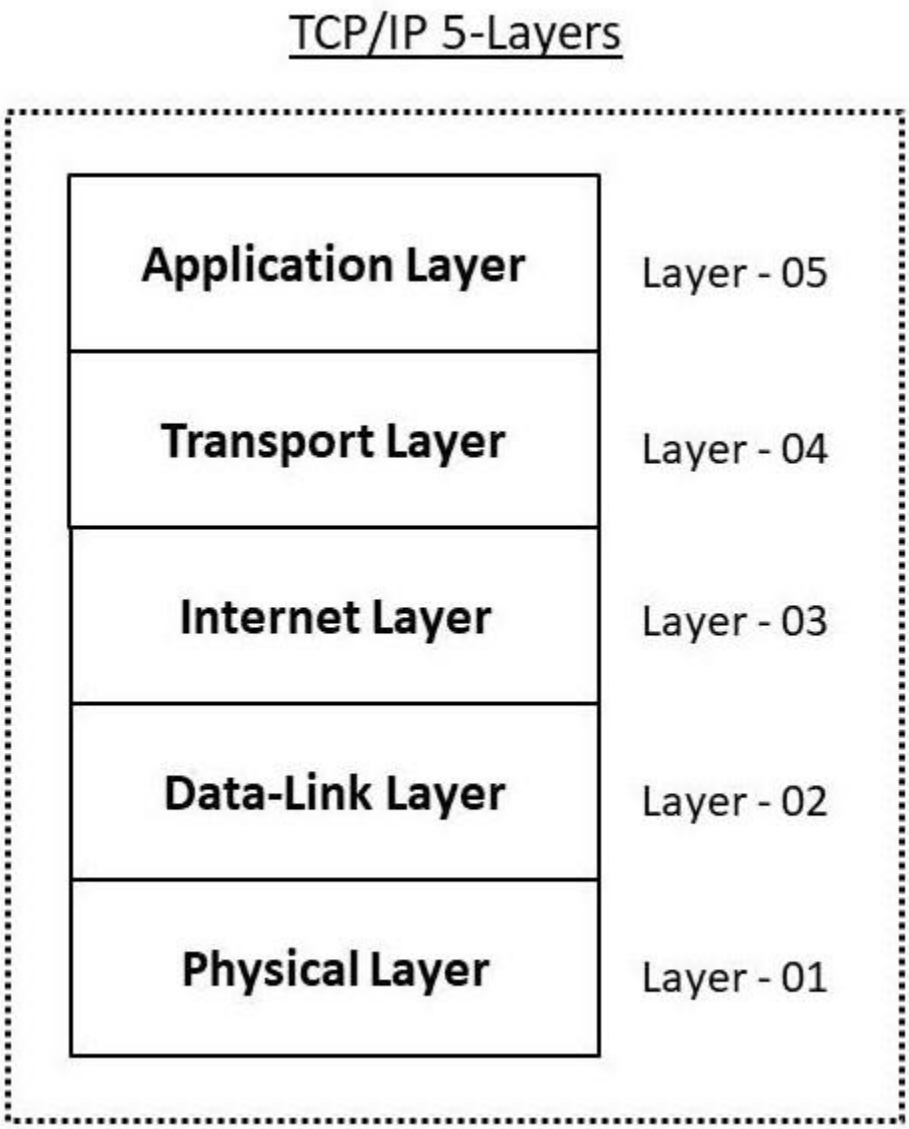


Figure (14) – TCP/IP Model

Explanation of protocols and devices map to different layers

Protocol and Device Mapping

The network design of BlueWave Industries needs to know protocol and device mapping in these models. Switches are used at Layer 2 (Data Link) with the MAC addresses being used to route the frames within each departmental network segment and the switching performance is microseconds to enable high-speed connectivity. The devices have dissimilar collision domain of each port, yet have a similar broadcast domain, which makes them an appropriate connection between the 79 user devices in the six departments, in their own subnets. Layer 3 switches or routers serve at the Network layer and operate on the use of the IP addresses to direct the packets between the departmental subnets that are being separated to enhance security. The router or Layer 3 switch employs as the default gateway of each subnet where appropriately controlled inter-departmental communication is permitted with the isolation of traffic by access control lists (ACLs) that specify which departments may communicate with others, e.g., all departments may communicate with servers in the IT Support Department, but direct communication between the HR and Sales and Marketing departments is prohibited due to confidentiality.

DNS and DHCP protocols are Application layer services, which include the necessary services in name resolution that changes the server names to IP addresses and also in IP address management that automotive the configuration of devices. TCP is used in the Transport layer to provide reliable delivery of important departmental communications such as email between the HR and employees, file transfers between the Finance and Management departments, and transactions to the database accessing business applications, using acknowledgment schemes and the automatic retransmission of lost packets. IP works on the Network layer, which gives the addressing structure which allows the subnetting scheme that is needed by BlueWave six departments, each department is assigned a separate subnet range within a private address space like 192.168.1.0/24 up to 192.168.6.0/24.

Application to BlueWave Industries

In the case of BlueWave industries, the layers model approach will enable an appropriate network design methodology and a methodological way of troubleshooting. Each of the Administration, Human Resources, Finance, Sales and Marketing, IT Support and Management departments will be made out of a separate Layer 3 subnet, allowing security policies to regulate the inter-departmental access, traffic management policies to prioritize the critical business application traffic, and usage monitoring policies that identify the trend of the departmental bandwidth usage. Each department has a Layer 2 switch to provide high speed connectivity to the local devices with Gigabit Ethernet (1 Gbps) ports to support workstations, printers, and 'VoIP' phones and Layer 3 routing to allow controlled inter-departmental communication over the default gateway router with firewall rules and access controls.

The knowledge of these models makes it possible to have the IT Support Department troubleshoot in a systematic way by isolating the problems to particular layers: physical layer problems include cable failure or NIC failures, data link layer problems include switch settings or MAC address clashes, network layer problems include routing problems or IP address problems, and application layer problems include DNS resolution problems or DHCP allocation problems. The layered configuration allows the adoption of corresponding security controls at every layer such as physical security of network closets that provide the storage of switches, MAC address filtering in the data link layer, firewall policies in the network layer and authentication in the application layer. The protocol mapping directs the choice of the right devices used in the infrastructure of BlueWave such as access layer switches to connect departments, distribution layer switches or routers to route between departments, server used to provide DHCP and DNS services, as well as access points used to provide the WLAN connectivity among other devices, and the configuration options to be used to ensure that the infrastructure can scale up to meet future needs in case of future expansion of the organization which can accommodate more than 79 users currently.

Assessment of Networking Principles and Protocol Effectiveness

Implementation of Core Networking Principles

Successful networks systems are based on principles that underlie communication and sharing of resources. In the case of BlueWave Industries Ltd., the layered communication model will allow troubleshooting to be done systematically using various layers that are separated by the functions (Imperva, 2024). When the Finance Department has connectivity problems, the IT Support department can systematically test physical connections, switch settings, IP addresses and applications instead of randomly looking at the full stack. Standardization of the protocols ensures the interoperability of the heterogeneous environment in BlueWave, where 79 users can communicate with Windows, macOS, and mobile devices without any compatibility issues.

Subnetting (network segmentation) improves performance and security since broadcast traffic is not allowed to travel outside the departments. The six departmental subnets of BlueWave allow queries in Finance database not to create broadcast traffic that would impact Sales and Marketing video conferencing and is effective in terms of traffic isolation. This segmentation establishes secure boundaries to access sensitive HR and Finance data not by other departments without assigned access points, which is a direct benefit to operational security.

Protocol Synergy Enabling Network Effectiveness

The effectiveness of networks arises due to the interaction of protocols that are not independent but work together. DHCP, DNS, and TCP/IP provide an automatic system with minimum technical expertise of the user, and it is centrally managed by IT. In the case of Administration recruiting new workers, DHCP assigns

IP addresses, subnet masks, and gateway data automatically, eliminating configuration errors and saving the IT department 75 percent of the workload that would have been required to do the same tasks with 79 users. DNS will allow connecting with resources with friendly names such as fileserver.bluewave.local whereas TCP will provide a stable delivery of data, which can be proven by operational efficiency.

Cloud-based business applications such as HTTP/HTTPS protocols can be used to foster secure web communication between all departments (Mosse Institute, 2022). The encryption of the banking operations of Finance and the employee data of HR offered by HTTPS ensures the safety of data transfers in case of sending and prevents the connection to fake systems provided by certificate authentication. SNMP can be used to oversee networks in a proactive manner, as it offers real-time monitoring of the performance of devices in six departments, which gives the IT Support team an opportunity to correct the situation before users can also notice the problem This will convert the reactive troubleshooting into proactive management which will be very beneficial in improving the level of reliability of the network.

Security Implementation and Quality of Service

Quality of Service (QoS) principles allow allocating the necessary resources to the applications according to their criticality. BlueWave network also focuses on the 79 VoIP phones ahead of general data traffic within the network which ensures clarity in the audio quality during congestion. Under ordinary circumstances, the executive video conferences of Management would suffer quality decline due to the transfer of huge files by Sales and Marketing but with appropriate prioritization, vital communications would be of consistent quality irrespective of what the other activities may be doing at the same time.

The effectiveness of security depends on the implementation of defense on a depth basis by a series of protocol layers. Web traffic is encrypted by HTTPS, remote

administration by SSH and firewall rules block malicious inter-subnet traffic. Such a multi-layered strategy implies that a violated security control does not provide total network access, which proves how the principles of security can be used to make the entire network efficient to ensure the safety of confidential financial and HR information of BlueWave.

Scalability and Long-Term Effectiveness

Any successful network is able to support growth without being fully redesigned. Hierarchy design and star topology of BlueWave enable the addition of users or departments through extension of the existing infrastructure. Flexibility in protocols allows adaptation - when BlueWave needs WAN connections with branch offices, the TCP/IP base can be extended to wide area connections without changing the applications. Scales of DNS and DHCP infrastructure are achieved by updating configuration instead of architecture by showing how protocol design can provide long-term capability as the organization grows.

This evaluation shows that networking principles which include, but are not limited to, layered architecture, standardization, segmentation, and security interact with protocols to form effective systems. The effectiveness of the network in BlueWave is the result of TCP/IP connectivity, DHCP/DNS configuration auto-service, HTTPS application security, SNMP management and QoS prioritization of critical network traffic. These protocols constitute a holistic ecosystem where the system functions are more than the capability of each component so that its functions can be reliable even while being flexible to future technological improvement.

Discussion Networking Devices and Operating Principles

Switches

Switches work in Layer 2 of the OSI model, performing the packet switching based on MAC address and sending the packet to the device in network segments.

They contain MAC address tables that use the device addresses to its physical port so as to forward it intelligently other than having to broadcast to all ports. In the case of BlueWave Industries, managed switches will be used to link the 79 user devices in departmental subnets and it will provide VLAN segmentation. Switches in the current generation have Spanning Tree Protocol that eliminates looping, port aggregation to handle more bandwidth and Quality of Service that prioritizes VoIP traffic as needed by the 79 VoIP phones.



Figure (15) – Switch Device

Routers

Routers work at Layer 3 and they use the IP addresses to deliver packets to other subnets. The central router is the default gateway of all the six departmental subnets of BlueWave Industries to allow controlled inter-departmental communication and enforce security by use of Access Control Lists. The router uses Network Address Translation (NAT) that enables 79 users to use a single public IP address to access the internet as well as the router uses private addressing internally.



Figure (16) - Router

Wireless Access Points

Wireless Access Points connect wireless devices to wired networks at the second layer, which is IEEE 802.11 standards. The seven WAPs of BlueWave will authenticate clients using WPA3 encryption, control radio channels to reduce interference, and bridge the wireless traffic to departmental VLANs. Newer WAPs can support multiple SSIDs of employee, guest and IoT networks with varying security policies and support 30-50 parallel connections per access point.

Firewalls

Firewalls block internet traffic according to pre-set rules and can work on the Layer 3 to Layer 7. In the case of BlueWave industries firewall is used to control inter-departmental traffic and this prevents interference with Finance and HR subnets. Deep packet inspection looks at the application data, which prevents the transfer of malware and it gives audit trails which can be used to investigate security.

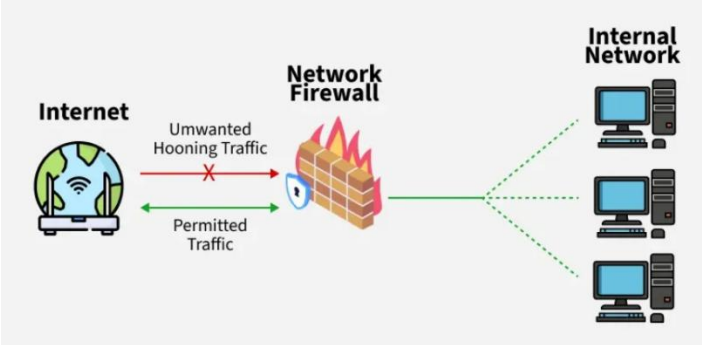


Figure (17) – How Firewall Works

Server Types

The DHCP Server, by automating IP address allocation, determines users of BlueWave numbering 79 users through the DORA process (Discovery, Offer, Request, Acknowledgment) and makes the process of IP allocation automatic and does not cause IP conflicts.

DNS Server is used to map a domain name to an IP address so that a user can use the “fileserver.bluewave.local” instead of remembering the IP addresses.

File server offers a centralized storage of documents via SMB/CIFS protocols with access control and a backup protection of data of an organization.

Database Server is a host that contains organized data to be used in business. The accounting systems used at the Finance Department of BlueWave are based on database servers and supported by ACID guarantees of handling financial transaction.

Web Server Web Server Web Server holds websites and web applications on HTTP/ HTTPS protocols and delivers content to the browsers visiting the applications of BlueWave intranet.

Mail server involves the use of SMTP to send emails and POP3/ IMAP to access emails, which is used to allow the 79 users to communicate with the spam filtering and malware scanning.

Discussion the interdependence of workstation hardware and relevant networking software

Hardware Parts used in workstations

The 79 workstations in BlueWave Industries include necessary hardware that allows them to be connected to the network. The Network Interface Card (NIC) has physical Ethernet or wireless access with its own MAC address used to identify Layer 2 clients and Gigabit Ethernet (1000 Mbps) support to access servers. Networking code and HTTPS encryption algorithms are executed in the processor (CPU). Protocol stacks and network buffers are found in memory (RAM). There are operating systems and networking software in storage devices and data which are stored in a cache. Components are connected with the motherboard using PCI Express slots and system buses.

Networking Software Components

Network operating systems (Windows, macOS, Linux) are used by workstations that provide TCP/IP protocol stack and device drivers operating the NIC hardware. TCP/IP stacks offer physical transmission to application protocols and this offers layers of networking. A device driver maps instructions given by the OS to NIC specific instructions. The client software is DHCP software, which automatically requests some IP configuration of the servers. DNS queries are used in accessing servers that convert hostnames into IP addresses. Internet browsers deploy HTTP/HTTPS that access business applications. Email clients support SMTP and IMAPs. VoIP software is used to enable voice coding and RTP to the 79 VoIP phones.

Interdependence Analysis

There is complete interdependence between workstation hardware and networking software. NICs also need device drivers to decode OS commands into

network messages, lack of which NICs will not be able to send data even though they are useful hardware. On the other hand, advanced networking software will not be able to operate the hardware of NIC that offers physical connectivity. The processing capabilities of the CPU also affect the networking performance- HTTPS encryption of the Finance Department banking system is a big drain in CPU cycles. Poor hardware will generate slow connection in spite of the quality of software.

Hardware timing precision is required of network protocol software. VoIP applications also demand NICs that provide packets within a few milliseconds to achieve a good quality of voice. Hardware with inadequate interrupt response is a source of packet delays and jitter, further worsening voice quality. In updating the operating systems, BlueWave should also provide new drivers that support new OS code and the current NIC hardware.

Application software relies on layers which are: HTTP/ HTTPS protocols rely on DNS resolution, which relies on TCP transmission, which relies on OS networking, which relies on NIC drivers, which relies on NIC hardware. The problems at one level block the functionality of the next level. One of the defective cables denies the Finance users access to the accounting software even after functional workstations and servers.

The case of 79 workstations in BlueWave requires compatibility planning in the acquisition of hardware, the choice of operating systems and applications installation. The IT Support Department shall make sure that the NIC hardware can handle the Gigabit speeds, the drivers are available to support the selected operating systems and the CPU/RAM is adequate to handle the networking and operation of applications simultaneously. This is a holistic solution that ensures a good connectivity assisting business activities in all six departments.

Designing an IP Addressing Scheme and Subnetting Structure for BlueWave Industries Ltd.

IP Addressing Requirement Analysis

Organizational Structure and Departmental Overview

BlueWave industries Ltd. has six departments that have different functioning needs and staff numbers. The staff of the organization consists of 79 employees at present which are distributed among Administration (18 users), Human Resources (10 users), Finance (12 users), Sales and Marketing (25 users), IT Support (8 users), and Management and Executive offices (6 users). Every department has its own network infrastructure that serves to carry on with the daily business without interfering with each other in terms of security segregation and effective resource allocation.

The Sales and marketing department is the biggest user base of 25 employees and they require a broad bandwidth of network to support the customer relationship management systems, video conferencing, transferring of large files to promote the product and collaboration tools. The Administration Department which has 18 users takes care of general business operations and inter-departmental coordination. The 12 users of the Finance Department deal with sensitive financial information that needs increased security and availability of database connectivity. The Human Resources has 10 employees who handle confidential information about employees. The smaller first-level IT Support Department of 8 technicians has a very important infrastructure operation that includes the process of network devices, servers, and security systems. The Department of Management and Executive Offices consisting of 6 senior personnel needs to have a secure communication in terms of strategic planning and making decisions.

Connection Network Devices Analysis

In addition to user workstations, the network at BlueWave industries should be capable of handling various connected devices that are required in the current day operations. All the 79 employees will need a workstation and a Voice over IP (VoIP) phone to have a minimum of 158 devices to be connected. VoIP offers affordable telephony and integrated communication which is tied to business applications. This two-gadgets necessitate has a strong effect on IP address designs and bandwidth planning.

There are eight shared printers located at strategic points within the departments and are referred to as network printing infrastructure. The sales and marketing division and administration division each need two printers because they have more documents and the finance, human resource, information technology support, and management divisions have one printer. These devices need to have fixed IP addresses so as to be accessible at any moment. The coverage of the wireless network is offered in the seven Wireless Access Points installed across the facility. The organization has two access points in Sales and Marketing that are necessary to support the activities of the department with a higher number of employees and mobile devices usage, whereas the rest of the departments have one apiece that support the use of laptops and BYOD policy.

Server Infrastructure Requirements and Justification

There are eleven vital servers that are part of the infrastructure in the server. The choice of server type balances between the cost optimization as well as the performance that is required based on the operational needs of BlueWave.

Servers of the IT Support Department (5 servers)

The DHCP Server is automated to allocate IP addresses to 79 users and other devices, which is why it requires specific hardware because it plays a critical role in aiming the network initialization. The implementation is cost-effective with virtual machine in small hardware (2 CPU cores, 4GB RAM, $800 capital cost) as DHCP is

minimal in processing. Reliability is an optimization of performance and not processing power.

DNS Server also offers name resolution services to all users and applications. Applications Justification Dedicated DNS server can support high-performance requirements of less than 100ms query response and redundancy requirements. It is estimated to cost around $1,200 to implement a physical server (4 cores, 8GB RAM) that has two network interfaces to offer fault tolerance. Optimization of performance by caching of most demanded records, which saved 80 per cent of query time on the internal resources.

Network Monitoring Server This server is a SNMP management software that monitors all the network devices, switches, routers, and servers. Justification also has proactive fault detection that will save down time worth much more than $2,500 in server costs (8 cores, 16GB RAM). The optimization of performance should be based on the adequate processing of real-time data collection of 240+ monitored objects and trend analysis in support of the capacity planning.

File server is used to store all documentations in a centralized place. Raid controller is necessary to protect data storage and its capacity of 4TB justifies a price of 3,500 dollars. Using SSD caching to optimize the performance of files that are frequently accessed speeds up the delivery of files, lessening network usage because they are cached locally with the backup and access control being centralized.

The 8TB Backup Server will be of worth in terms of cost since it enables the organization to recover data that is significantly worth much more than the equipment cost. By scheduling performance optimization backups at off-hours, the backup will not affect the business.

Servers (2 servers) of the finance department

Database Servers containing accounting and financial programs warrant high-end hardware (6,000 each) that contains a redundant power supply, RAID disk storage system and 32GB of RAM to support simultaneous transaction processing.

Database indexing and query optimization are used to achieve performance optimization to ensure financial reports can be processed in sub-second response times. Cost justification involves avoiding delays in financial transactions and making financial visibility available in real-time to make decisions.

Other Departmental Servers (4 servers)

Each of Sales & Marketing, Administration, HR and Management has application or file servers (both cost the same, 2000) of a department specific software. The mid-range specifications (4 cores, 16GB RAM, 1TB storage) are found to be balanced on the cost and performance requirement of the office use. The virtualization technology enables more departmental servers to be put on the physical machines that are few which make the overall cost to be reduced to about 5000 dollars whilst logical separation is still maintained.

Reckoning initial capital expenditure vs. operational efficiency, low cost of downtime and the capability of the business to do business much the server infrastructure costs about 27,000 on the whole. Performance optimization is the optimization of resources based on the applications that are transaction-intensive (Finance databases) and the less intensive applications are handled through cost-effective solutions (DHCP, DNS).

Present and Future Host Requirement

The present operational needs amount to 184 interconnected equipment 79 workstations, 79 VoIP phones, 8 printers, 7 wireless access points, 11 servers, network infrastructure equipment. The network design should be able to support growth. Best practice in the industry suggests 20-30 percent growth capacity. The growth trend of BlueWave will require the inclusion of 30 percent more capacity, which will result in a growth of about 24 more employees along with corresponding increases in equipment.

The entire estimated needs are 240 devices that include the present equipment and expansion potential. The growth forecasts of departments are different depending on the nature of operations practiced. There is a high-capacity

demand in Sales & Marketing of 17 more hosts than the current requirement, which will amount to 72 hosts. Administration will require 12 extra hosts making it 52. Finance has 8 others, to a total of 36 hosts. HR grows by 7 to 30 total hosts. IT support increases to 30 hosts to support its services. The management will need small increment of 5 more hosts numbering to 20.

Summary of IP Addressing Requirement

According to the analysis, BlueWave Industries is in need of IP addressing that can serve up to 240 total hosts that include six departments with varying needs ranging between 20 hosts (Management) and 72 hosts (Sales and Marketing). The scheme should use private IP addressing that is in agreement with the RFC 1918 specifications, apply Variable Length Subnet Masking (VLSM) to allow effective use, give different subnets to allow security segregation and traffic control, support VLAN implementation to allow logical separation of a shared infrastructure, and provide Quality of Service that can support business critical traffic, and allow future expansion without the need to redesign the scheme.

Department	Users	Printers	VoIP Phones	Wireless APs	Servers	Future Growth (30%)	Total Hosts Required	Recommended Subnet
Sales & Marketing	25	2	25	2	1	17	**72**	/25 (126 hosts)
Administration	18	2	18	1	1	12	**52**	/26 (62 hosts)
Finance	12	1	12	1	2	8	**36**	/26 (62 hosts)
Human Resources	10	1	10	1	1	7	**30**	/27 (30 hosts)
IT Support	8	1	8	1	5	7	**30**	/27 (30 hosts)
Management	6	1	6	1	1	5	**20**	/27 (30 hosts)
TOTAL	**79**	**8**	**79**	**7**	**11**	**56**	**240**	-

Figure (18) - Requirements Analysis Table

Purpose and Importance of Subnetting

Subnetting is an essential network design concept, which is currently used in the organizational networks and the large networks are subdivided into small networks known as subnets. Subnetting plays several very important roles in modern-day business settings such as BlueWave Industries, outnumbering the involvement in addressing organization. The contemporary organizations are

experiencing more pressures such as cyberattacks that are aimed at targeting sensitive information, network overloading by applications that consume bandwidth such as video conferencing and cloud computing, and the fast growth of organizations that demand dynamic infrastructure. Subnetting is a direct response to these issues in that it offers enhanced security because of network segmentation, traffic control because of reduced broadcast domains and scalability because of modular network architecture. In the case of BlueWave Industries where six departments deal with varying data sensitivity categories and every department has different application requirements, subnetting will convert the flat and vulnerable network to a structured and secure network that is efficient to support the business operation and allows further expansion in the future.

Security Advantages Network Segmentation

Subnetting gives the separation of the network into departments that would be autonomous of broadcasting areas and security locations. In the case of BlueWave Industries, it implies that the Finance department with sensitive banking data is on subnet 192.168.0.192/26 with Sales and Marketing being on a different subnet 192.168.0.0/25. The router has Access Control Lists (ACLs) that regulate the communication between departments to ensure that financial records are not accessed by unauthorized individuals. Under non subnetting, any workstation attacked within Sales and Marketing might have a chance of gaining access into the database server of Finance department in the same flat network. In subnet partition, the router will become the gate of security where it will stop the unauthorized connections and allow authorized access requests. In the same way, the confidential employee information of HR Department that is in subnet 192.168.1.0/27 is not accessible to casual browsing by other departments. Hence, data protection regulations are adhered to.

Traffic Control and Performance Maximum

Subnetting enhances performance in the network by limiting broadband messages to departmental lines. Using non-subnetting, all the 79 users and all the 103 devices would belong to the same broadcast domain wherein all broadcast packets are received by all the devices, wasting bandwidth and processing resources. Broadcast storms initiated by faulty equipment or wrongly set up applications would affect the organization as a whole. Six individual subnets Sales and marketing with 25 users who add broadcast traffic that is used in file sharing and printer discovery are not going to affect the database queries of the Finance Department. This network isolation guarantees the same performance of applications to departments, which is especially significant to Finance due to time-sensitive financial reporting and Sales and Marketing due to video conferences to customers.

Scalability

The subnetting design allows maximum growth of the organization without redesign of infrastructure. The current 79 users that BlueWave has can be expanded to about 103 users under the current subnets. Sales and Marketing subnet of 126 available hosts will support current 72 hosts and 54 hosts that will be added in future without any modification. In case the departments ultimately surpass the capacity set aside, it is possible to establish some new subnets using the available 192.168.x.x address space without affecting the operations of current subnets. Modular subnet design provides the option of adding new department or branch offices since it can assign new subnet ranges without altering the current subnet structure.

Subnetting gives flexibility when the organization is to change. In case BlueWave develops a new department, the IT Support team only has to configure a new subnet using unused address space, configure DHCP pool, and matching VLAN,

and modify firewall rules, without having to alter departmental settings. This is scaled out, which is not feasible in the case of flat network architecture where address exhaustion necessitates full redesign that impacts users at once.

Administrative and Operational Benefits

Subnetting makes it easier to administer the network and troubleshooting to the IT Support Department. Well defined subnet boundaries make finding pertinent departments easy as per the IP address range. Once users are reporting connectivity problems, the IT team knows automatically which subnet, VLAN, and switch to check. The DHCP servers, when set to allocate address pools per subnet, automatically assign the appropriate parameters of the network according to the area of the user and this way, there is no error in configuration.

Policies such as bandwidth quota, content blocking, or backup times can be implemented at the subnet level, instead of per host, and make configuration a much less complicated task with 240 total hosts. The use of subnet-based management allows distributing some administrative functions to departmental IT contacts and still retains the overall control. Standardized subnet architecture encourages paperwork, educating new IT personnel, and standardized network practices throughout the organization.

Development of an appropriate private IP addressing scheme

Selection and Justification of IP Range

There is a 192.168.0.0/16 address space as indicated in RFC 1918 that BlueWave Industries uses due to various strategic reasons. First, it has sufficient address space to meet the existing demands (240 hosts) and large future capacity beyond 65,000 possible addresses. This massive ceiling space is enough to house possible branch offices, new departments, guest networks, internet of things, and even organizational expansion decades without exhausting the address issue.

Second, the private addressing removes the internet routing conflicts and security threats that come with the use of public IP addresses. The public IP addresses are routable in the globe and thus any device on the Internet may try to make connections to the internal resources of BlueWave. Non-routable addresses on the public Internet are inherently secure as they are isolated, and they are private. 192.168.x.x addresses cannot be directly accessed through external attackers, and they have to obtain access to the perimeter firewall or VPN gateway.

Third, the 192.168.x.x range is an industry standard of small-to-medium businesses to guarantee compatibility with standard networking equipment and familiarity of the administrator. The IT employees moving to BlueWave have probable experience with 192.168.x.x networks and this cuts down the time spent on training. Fourth, the internet gateway has Network Address Translation (NAT) so that all users (79) can use a single public IP address to access the internet, which will save on cost to a great extent. The cost of a single IP address in many different locations is high with the ISPs, whereas NAT enables unlimited access to internal devices within a single public address.

Subnetting VLSM Methodology

Variable Length Subnet Masking (VLSM) is effective in addressing allocation of addresses since it allocates subnets on a departmental basis and not on an identical basis. The conventional fixed-length subnetting would allocate all the six departments /27 subnets (30 hosts each) and waste the addresses of smaller departments and under serve the larger ones. VLSM enables Sales & Marketing receiving /25 (126 hosts) to get 72 hosts that are required and Management receiving /27 (30 hosts) to get 20 required hosts so that the wastage is minimized.

VLSM calculation process includes the first step of ranking the ordering departments by the size Sales and Marketing (72 hosts) is the largest, then comes Administrations (52), Finances (36), HR (30), IT Support (30), and Management (20) are the smaller ones. This is because when the largest department is started, there is adequate address space left to other departments. Subnet size of each

department is computed based on formula: 2^n -2 [?] required hosts where n is equal to the host bits. The -2 takes into consideration those network address and broadcast address that are not available to hosts.

Subnetting Calculations

For Sales & Marketing Department

Requirements Analysis:

Users: 25

Additional devices: 25 VoIP phones + 2 printers + 2 wireless APs + 1 server = 30 devices

Total current needs: 55 hosts

Future growth (30%): 17 additional hosts

Total required: 72 hosts

Subnet Calculation:

Formula: 2^n - 2 ≥ 72 (n = host bits)

Testing: 2^6 - 2 = 62 hosts (insufficient)

Testing: 2^7 - 2 = 126 hosts(sufficient)

Host bits needed: 7

Network bits: 32 - 7 = 25

Subnet Mask: /25 or 255.255.255.128

Binary subnet mask: 11111111.11111111.11111111.10000000

Address Allocation:

Network Address: 192.168.0.0

First Usable IP: 192.168.0.1 (default gateway)

Last Usable IP: 192.168.0.126

Broadcast Address: 192.168.0.127

Total available hosts: 126

Hosts utilized: 72

Remaining capacity: 54 hosts (43%)

For Administration Department

Requirements Analysis:

Users: 18

Additional devices: 18 VoIP phones + 2 printers + 1 wireless AP + 1 server = 22 devices

Total current needs: 40 hosts

Future growth (30%): 12 additional hosts

Total required: 52 hosts

Subnet Calculation:

Formula: $2^n - 2 \geq 52$

Testing: $2^5 - 2 = 30$ hosts (insufficient)

Testing: $2^6 - 2 = 62$ hosts (sufficient)

Host bits needed: 6

Network bits: 32 - 6 = 26

Subnet Mask: /26 or 255.255.255.192

Binary subnet mask: 11111111.11111111.11111111.11000000

Address Allocation:

Network Address: 192.168.0.128

First Usable IP: 192.168.0.129 (default gateway)

Last Usable IP: 192.168.0.190

Broadcast Address: 192.168.0.191

Total available hosts: 62

Hosts utilized: 52

Remaining capacity: 10 hosts (16%)

For Finance Department

Requirements Analysis:

Users: 12

Additional devices: 12 VoIP phones + 1 printer + 1 wireless AP + 2 servers (financial databases) = 16 devices

Total current needs: 28 hosts

Future growth (30%): 8 additional hosts

Total required: 36 hosts

Subnet Calculation:

Formula: 2^n - 2 ≥ 36

Testing: 2^5 - 2 = 30 hosts (insufficient)

Testing: 2^6 - 2 = 62 hosts (sufficient)

Host bits needed: 6

Network bits: 32 - 6 = 26

Subnet Mask: /26 or 255.255.255.192

Address Allocation:

Network Address: 192.168.0.192

First Usable IP: 192.168.0.193 (default gateway)

Last Usable IP: 192.168.0.254

Broadcast Address: 192.168.0.255

Total available hosts: 62

Hosts utilized: 36

Remaining capacity: 26 hosts (42%)

For Human Resources Department

Requirements Analysis:

Users: 10

Additional devices: 10 VoIP phones + 1 printer + 1 wireless AP + 1 server = 13 devices

Total current needs: 23 hosts

Future growth (30%): 7 additional hosts

Total required: 30 hosts

Subnet Calculation:

Formula: 2^n - 2 ≥ 30

Testing: 2^5 - 2 = 30 hosts (exact fit)

Host bits needed: 5

Network bits: 32 - 5 = 27

Subnet Mask: /27 or 255.255.255.224

Binary subnet mask: 11111111.11111111.11111111.11100000

Address Allocation:

Network Address: 192.168.1.0

First Usable IP: 192.168.1.1 (default gateway)

Last Usable IP: 192.168.1.30

Broadcast Address: 192.168.1.31

Total available hosts: 30

Hosts utilized: 30

Remaining capacity: 0 hosts (100% utilization - may need monitoring)

For IT Support Department

Requirements Analysis:

Users: 8

Additional devices: 8 VoIP phones + 1 printer + 1 wireless AP + 5 servers (DHCP, DNS, monitoring, file, backup) = 15 devices

Total current needs: 23 hosts

Future growth (30%): 7 additional hosts

Total required: 30 hosts

Subnet Calculation:

Formula: 2^n - 2 ≥ 30

Testing: 2^5 - 2 = 30 hosts (exact fit)

Host bits needed: 5

Network bits: 32 - 5 = 27

Subnet Mask: /27 or 255.255.255.224

Address Allocation:

Network Address: 192.168.1.32

First Usable IP: 192.168.1.33 (default gateway)

Last Usable IP: 192.168.1.62

Broadcast Address: 192.168.1.63

Total available hosts: 30

Hosts utilized: 30

Remaining capacity: 0 hosts (100% utilization)

For Management & Executive Offices Department

Requirements Analysis:

Users: 6

Additional devices: 6 VoIP phones + 1 printer + 1 wireless AP + 1 server = 9 devices

Total current needs: 15 hosts

Future growth (30%): 5 additional hosts

Total required: 20 hosts

Subnet Calculation:

Formula: 2^n - 2 ≥ 20

Testing: 2^4 - 2 = 14 hosts (insufficient)

Testing: 2^5 - 2 = 30 hosts ✓ (sufficient)

Host bits needed: 5

Network bits: 32 - 5 = 27

Subnet Mask: /27 or 255.255.255.224

Address Allocation:

Network Address: 192.168.1.64

First Usable IP: 192.168.1.65 (default gateway)

Last Usable IP: 192.168.1.94

Broadcast Address: 192.168.1.95

Total available hosts: 30

Hosts utilized: 20

Remaining capacity: 10 hosts (33%)

Calculation Summary and Validation

VLSM subnetting scheme with a total of 340 IP addresses (126+ 62 + 62 + 30 + 30 + 30) can support 240 needed hosts which is a 71% efficiency. Subnets have no interruptions, and they use the addresses space of 192.168.0.0 to 192.168.1.95. There is no overlap of subnets and the growth capacity is met accordingly to the requirements of the departments. The calculations exhibit acceptable binary subnet mathematics, network and broadcast address identification and sound determination of host range of each departmental subnet.

Department	Network Address	Subnet Mask	CIDR	First Usable IP	Last Usable IP	Broadcast Address	Default Gateway	Total Available Hosts	Hosts Used	Remaining Capacity
Sales & Marketing	192.168.0.0	255.255.255.128	/25	192.168.0.1	192.168.0.126	192.168.0.127	192.168.0.1	126	72	54
Administration	192.168.0.128	255.255.255.192	/26	192.168.0.129	192.168.0.190	192.168.0.191	192.168.0.129	62	52	10
Finance	192.168.0.192	255.255.255.192	/26	192.168.0.193	192.168.0.254	192.168.0.255	192.168.0.193	62	36	26
Human Resources	192.168.1.0	255.255.255.224	/27	192.168.1.1	192.168.1.30	192.168.1.31	192.168.1.1	30	30	0
IT Support	192.168.1.32	255.255.255.224	/27	192.168.1.33	192.168.1.62	192.168.1.63	192.168.1.33	30	30	0
Management	192.168.1.64	255.255.255.224	/27	192.168.1.65	192.168.1.94	192.168.1.95	192.168.1.65	30	20	10

Figure (19) – IP Addressing Table

The addressing scheme has shown to be efficiently used with 240 required devices being served as of only 340 allocated addresses (192.168.0.0 through 192.168.1.95) which is 57 percent overall efficiency and can still grow. Such a balance will help avoid waste of addresses at the same time as making sure that departments do not spend money immediately on divisional requirements that entail disruptive growth. Subnet boundaries are based on convenient binary boundaries which makes them easier to manage. Network addresses (192.168.0.0, 192.168.0.128, 192.168.0.192, etc.) apply round numbers which can be easily remembered by the IT personnel. The allocation of addresses without gaps between neighboring addresses prevents difficulty in address fragmentation.

Implementation Considerations

The lower section of each subnet will be used as static IP addresses to infrastructure devices (routers, switches, printers, servers, wireless access points), and upper ranges will be used as DHCP pools by workstations and VoIP phones. As an example, in Finance subnet 192.168.0.192/26, gateway is allocated as .193, printer and servers are allocated their static address for .194-.200, and finally VoIP phones and workstations are allocated in the DHCP pool as .201-254.

VLAN IDs map directly to subnets enabling intuitive troubleshooting: VLAN 10 for Sales & Marketing (192.168.0.0/25), VLAN 20 for Administration (192.168.0.128/26), VLAN 30 for Finance (192.168.0.192/26), VLAN 40 for HR (192.168.1.0/27), VLAN 50 for IT Support (192.168.1.32/27), and VLAN 60 for

Management (192.168.1.64/27). This consistent mapping simplifies configuration and troubleshooting across network infrastructure.

Using critical reflection to evaluate the network system, security requirements and quality of services needed for enhance the valid conclusion

Network System Evaluation

The proposed network design will support the needs of BlueWave Industries with the help of a star topology that will provide a centralized management system, as well as VLSM subnetting that will allow managing the network efficiently and VLAN segmentation that will help increase the level of security. The system proves to be efficient in resource utilization with 240 devices supported on six subnets provided with less than 192.168.0.0-192.168.1.95 address range (less than 1% of available private space), giving much room to scale in the future. There are however single points of failure in the central router/switch which need to be considered with redundancy devices or fast replacement mechanisms. The eight technicians in the IT Support Department are able to handle the infrastructure by using centralized SNMP monitoring, automate DHCP system, and design in hierarchy, but documentation standards and change management procedures should be implemented to ensure integrity of the systems are maintained as complexity increases.

Security Requirements Assessment

The security needs of BlueWave Industries in each department depend on the sensitivity of data. The Finance Department that oversees banking credentials, payroll, and financial records must have the highest security classification and be able to access it only with HTTPS, files must be encrypted and must not inter-departmental access thus the firewall rules must be strict and must have advanced logging, in order to comply with audit requirements. The Human Resources that

manage confidential employee records, performance reviews and medical information should have the same high-security controls and have limited access to HR subnet of 192.168.1.0/27. Sales & Marketing and Administration demand medium security that secures business correspondence and client data by means of HTTPS encryption and conventional firewall regulations. ACLs can be used to implement various security policies by department using the subnet-based architecture, where the Finance and HR subnets can be set to block unsolicited incoming connections and permit outbound traffic and access to central servers.

Quality of Service Analysis

The implementation of the Quality of Service is essential to support the 79 VoIP phones in BlueWave that need the constant and low-latency delivery of packets. The priority of traffic given by QoS policies must be as follows: (1) VoIP with the highest priority that ensures clear audio during congestion, (2) database transactions to the Finance Department finance-related applications, (3) business-critical web applications such as CRM and accounting system, (4) general data traffic such file transfers and email, and (5) internet browsing with the lowest priority. This prioritization is important in ensuring that business critical applications remain available in terms of performance when the Sales and Marketing Department is accessing 25 users at a time to carry out its applications such as conducting video conferences or transferring large files. QoS implementation involves the setup of switches and routers to check the markings on packets (Diff Serv Code Points) and route high-priority packets over other low-priority packets in times of congestion. Bandwidth reservation of 128 kbps per VoIP phone is to guarantee the voice quality irrespective of the levels of data traffic.

The Valid Conclusions

The network configuration is effective in combining the present needs of BlueWave industries that has 79 users and six departments and 30 percent capacity in the growth of the network through proper subnetting. VLAN Star topology is a topology that provides cost-effectiveness, security, performance, and manageability.

The proper documentation of IP allocations, the stable use of QoS policies, routine security audits of the ACL configurations, and active monitoring by the use of SNMP are considered critical success factors. To improve it, it is recommended to install redundancy of core networking devices that avoid single points of failure, formal change management procedures of configuration changes, install intrusion detection systems to monitor Finance and HR subnets and block attempts of unauthorized access, and conduct regular reviews of the capacity to ensure that subnets are not overutilized (70-85) and trigger infrastructure planning before they become overutilized. The solution suggested offers a scalable, secure, and efficient network backbone to enable BlueWave Industries in its business operations and future organizational development.

Conclusion

This report has provided a detailed network design solution to the BlueWave Industries Ltd. to meet its immediate needs in terms of 79 users in six departments as well as cater to their future needs. Part I has explored some basic concepts of networking such as types of networks, topologies, protocols and OSI/TCP-IP model, which has made the choice of star topology and proper protocols to use to be business effective. Part II created a workable scheme of IP addresses based on VLSM subnetting assigning addresses effectively out of the 192.168.0.0/16 range of personal addresses and yet avoiding overlapping of addresses in the security division and scalability as well.

The solution proposed has managed to strike a balance between cost-effectiveness, performance, security as well as manageability. Star topology offers centralized control of the IT Support Department, the separation of sensitive Finance and HR data into VLANs, and the implementation of the QoS will provide stable VoIP communications. The subnetting design supports a 30 percent growth capacity without having to redesign the infrastructure to support further expansion, so that BlueWave Industries can continue to expand its business and can sustain the

operational efficiency and security levels befitting business contexts of the current times.

References

Anon., 2023. *TheLaw.Institute.* [Online]
Available at: https://thelaw.institute/cyberspace-technology-and-social-issues/understanding-networking-protocols-telnet-ftp-smtp-http/
[Accessed 8 12 2025].

Anon., 2025. *GeeksforGeeks.* [Online]
Available at: https://www.geeksforgeeks.org/computer-networks/overview-of-personal-area-network-pan/
[Accessed 8 12 2025].

Anon., 2025. *GeeksforGeeks.* [Online]
Available at: https://www.geeksforgeeks.org/computer-networks/simple-network-management-protocol-snmp/
[Accessed 8 12 2025].

Anon., 2025. *geeksforgeeks.org.* [Online]
Available at: https://www.geeksforgeeks.org/computer-networks/network-time-protocol-ntp/
[Accessed 8 12 2025].

Anon., 2025. *geeksforgeeks.org.* [Online]
Available at: https://www.geeksforgeeks.org/computer-networks/arp-protocol/
[Accessed 8 12 2025].

Anon., 2025. *geeksforgeeks.org.* [Online]
Available at: https://www.geeksforgeeks.org/computer-networks/open-systems-interconnection-model-osi/
[Accessed 8 12 2025].

CloudFlare, n.d. *cloudflare.com.* [Online]
Available at: https://www.cloudflare.com/learning/access-management/what-is-ssh/
[Accessed 8 12 2025].

Committee, T. L. S., 2024. *standards.ieee.org.* [Online]
Available at: https://standards.ieee.org/wp-content/uploads/2024/02/802.3-2022_errata.pdf
[Accessed 8 12 2025].

Cox, S., 2022. *controldesign.com.* [Online]
Available at: https://www.controldesign.com/connections/data-acquisition-

monitoring/article/21438562/bus-ring-star-or-mesh-which-topology-is-right-for-you
[Accessed 8 12 2025].

Fritzler, L., 2024. *It Planet.* [Online]
Available at: https://blog.it-planet.com/en/network-types-decrypted-your-comprehensive-guide-to-wan-lan-wlan-vlan-can-gan-man-pan-and-vpn/
[Accessed 8 12 2025].

GeeksforGeeks, 2025. *GeeksforGeeks.* [Online]
Available at: https://www.geeksforgeeks.org/computer-networks/types-of-area-networks-lan-man-and-wan/
[Accessed 8 12 2025].

IPCisco, n.d. *ipcisco.com.* [Online]
Available at: https://ipcisco.com/lesson/network-topologies/
[Accessed 8 12 2025].

IPCisco, n.d. *IPCisco.com.* [Online]
Available at: https://ipcisco.com/lesson/types-of-networks/
[Accessed 8 12 2025].

Kapoor, A., 2025. *Simplilearn.* [Online]
Available at: https://www.simplilearn.com/tutorials/networking-tutorial/importance-of-types-of-networks-lan-man-wan
[Accessed 8 12 2025].

Whittaker, G., 2024. *linuxjournal.com.* [Online]
Available at: https://www.linuxjournal.com/content/unlocking-linux-networking-essential-insights-tcpip-dns-dhcp-and-routing
[Accessed 8 12 2025].